

MODULE 1- Properties of Matter

Properties of matter

Inquiry question: How do the properties of substances help us to classify and separate them?

- Explore homogeneous mixtures and heterogeneous mixtures through practical investigations:
 - using separation techniques based on physical properties
 - calculating percentage composition by weight of component elements and/or compounds

- Investigate the nomenclature of inorganic substances using International Union of Pure and Applied Chemistry (IUPAC) naming conventions

- Classify the elements based on their properties and position in the periodic table through:
 - physical properties
 - chemical properties

Atomic structure and atomic mass

Inquiry question: Why are atoms of elements different from one another?

- Investigate the basic structure of stable and unstable isotopes by examining:
 - their position in the periodic table
 - the distribution of electrons, protons and neutrons in the atom
 - representation of the symbol, atomic number & mass number (nucleon number)

- Model the atom's discrete energy levels, including electronic configuration and spdf notation
- Calculate the relative atomic mass from isotopic composition

- Investigate energy levels in atoms and ions through:
 - collecting primary data from a flame test using different ionic solutions of metals
 - examining spectral evidence for the Bohr model and introducing the Schrödinger model

- Investigate the properties of unstable isotopes using natural and human-made radioisotopes as examples, including but not limited to:
 - types of radiation
 - types of balanced nuclear reactions

Periodicity

Inquiry question: Are there patterns in the properties of elements?

- Demonstrate, explain and predict the relationships in the observable trends in the physical and chemical properties of elements in periods and groups in the periodic table, including but not limited to:
 - state of matter at room temperature
 - electronic configurations and atomic radii
 - first ionisation energy and electronegativity
 - reactivity with water

Bonding

Inquiry question: What binds atoms together in elements and compounds?

- Investigate the role of electronegativity in determining the ionic or covalent nature of bonds between atoms

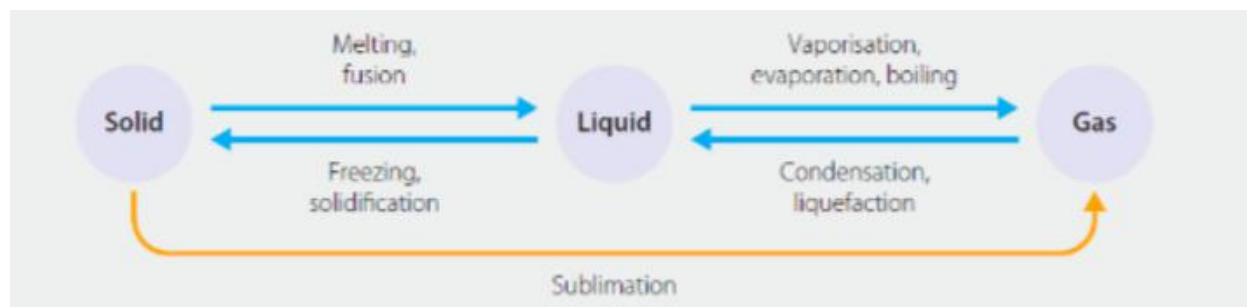
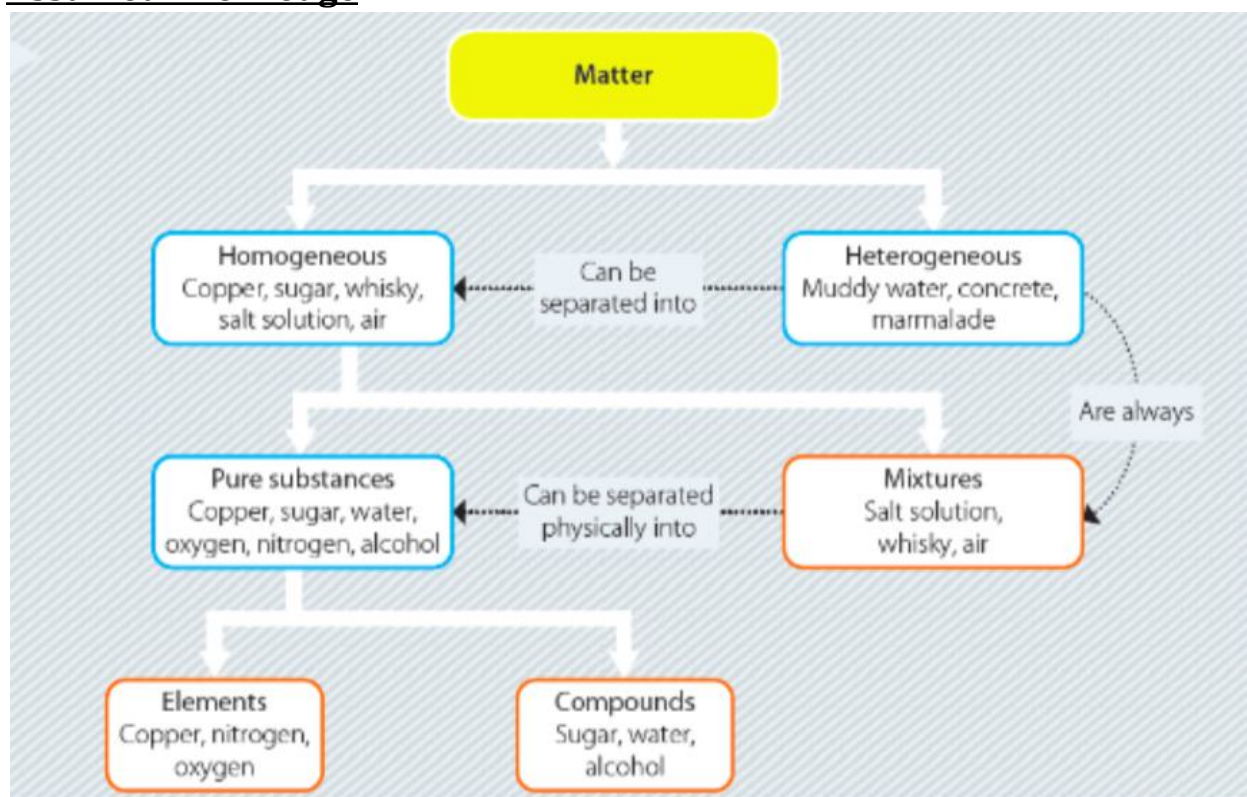
- Investigate the differences between ionic and covalent compounds through:
 - using nomenclature, valency and chemical formulae (including Lewis dot diagrams)
 - examining the spectrum of bonds between atoms with varying degrees of polarity with respect to their constituent elements' positions on the periodic table
 - modelling the shapes of molecular substances

- Investigate elements that possess the physical property of allotropy

- Investigate the different chemical structures of atoms and elements, including but not limited to:
 - ionic networks
 - covalent networks (including diamond and silicon dioxide)
 - covalent molecular
 - metallic structure

- Explore the similarities and differences between the nature of intermolecular and intramolecular bonds and the strength of the forces associated with each, in order to explain the:
 - physical properties of elements
 - physical properties of compounds

Assumed knowledge



Chemical change	Physical change
At least one new substance is formed	No new substance formed
Difficult to reverse	Easy to reverse
A large amount of energy	Limited amount of energy
Change in colour/temperature	Change in appearance
Chemical properties: Reactivity, flammability, acidity, toxicity	Physical properties: colour, magnetism, particle size, melting and boiling point, density

Separation techniques based on physical properties

- **Solution:** A homogenous mixture in which dispersed particles are so small, that they don't settle
- **Solute:** Substance that dissolves
- **Solvent:** Solution the substance dissolves in
- **Suspension:** Particles dispersed in a liquid

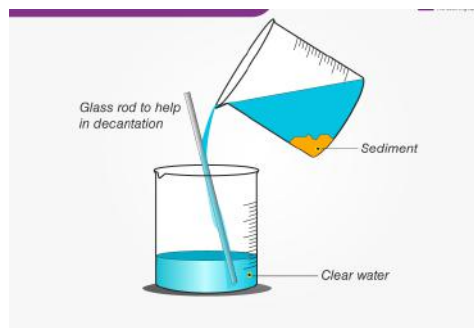
Sieving

- Used to separate solids of different particle size
- E.g. fine sand from gravel



Sedimentation and decanting

- Sedimentation is when heavier components of a mixture settle to the bottom, due to gravity (suspension). E.g. mud and water in muddy water
- Decanting is for separating two liquids with different densities by removing the layer of liquid at the top from the layer of solid/liquid. E.g. dirt and water



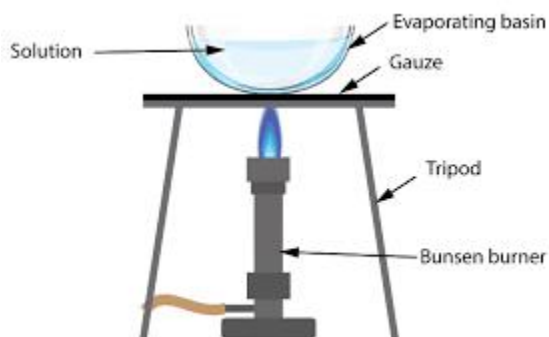
Filtration

- Used to separate solids from liquids based on differences in the states and particle sizes
 - Insoluble solid and liquid
 - It uses semi-permeable filter paper that has pores that are small enough for liquids but not solids
- For example, filtering salt and water in seawater



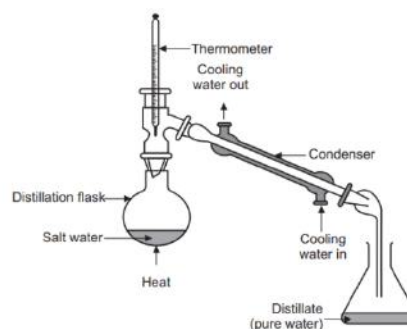
Evaporation

- Used to separate fluids or solutes from a solution based on differences in boiling point
- Liquid usually has a lower boiling point than the solid
 - Can only retrieve one substance
- E.g. salt from water in saltwater



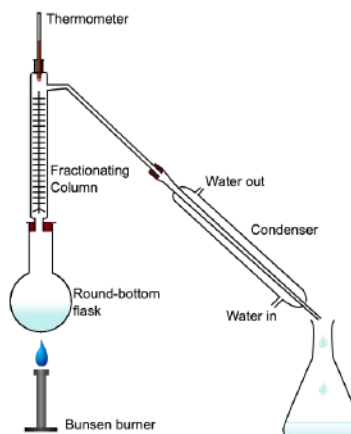
Distillation

- Used to separate **fluids** or **solutes** from a **solution** based on differences in boiling point and can have two or more liquids
 - A mixture of liquids is boiled with vapour formed, then condensed back into liquid
 - **Greater** difference than 25°C



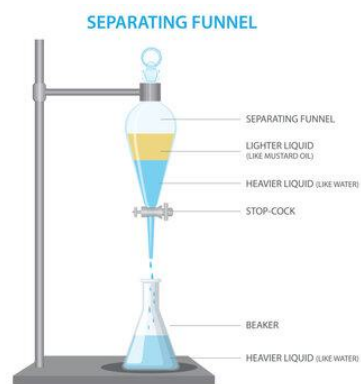
Fractional distillation

- Used to separate **fluids** or **solutes** from a **solution** based on **less** difference than 25°C in boiling point and can have two or more liquids
- **Fractional column:** Allows for repeated vaporisation and condensation up and down the column which means liquids with lower boiling points emerge as the distillate first
 - Distillate (original substance) will not be pure but both substances will be retrieved
- E.g. separating components of crude oil



Separating funnel

- Used to separate two **immiscible** (don't dissolve) liquids based on density
 - The **bottom** layer is dispersed through the **funnel** and collected in a beaker (denser solvent = bottom layer)
 - Only a small sample of one of the substances is retrieved but it will be pure
- E.g. oil and water (water is more dense)



Gravimetric analysis

- Determining the quantities (masses) of substances present in a sample

$$\% \text{ weight} = \frac{\text{Mass of component in sample (g)}}{\text{Total mass of sample (g)}} \times 100$$

Atomic structure

- The nucleus is made up of protons and neutrons and is surrounded by an electron cloud
 - Nucleons used to describe protons and neutrons
- **Electron shells:** Energy levels where electrons of an atom exist

Particle	Symbol	Relative mass	Relative charge
Electron	e	1/2000	-1
Proton	p	1	+1
Neutron	n	1	0

Isotopes

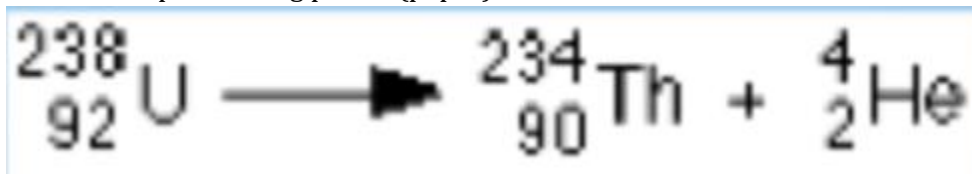
- Different forms of the same element (different number of neutrons) - Same chemical properties - Similar physical properties	$\begin{matrix} A \\ Z \end{matrix} X$ A = Atomic mass (protons + neutrons) Z = Atomic number (protons) X = Element
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For stable isotopes:

- $Z < 20$, n:p $\rightarrow$ 1:1
- $Z = 50$, n:p $\rightarrow$ 1:1.3
- $Z = 80$, n:p $\rightarrow$ 1:1.5
- $Z > 83$, no stable isotopes

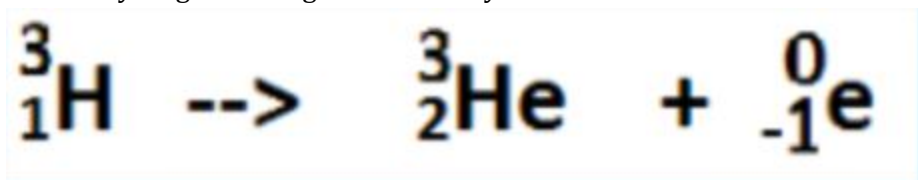
Alpha Decay (a)

- Nucleus ejects alpha particles (clusters of two protons and two neutrons)
- Only occurs in elements with a mass greater than 100
- When an element undergoes alpha decay, the atomic number decreases by 2 and the mass decreases by 4
- Equivalent to a helium nucleus with a charge of +2
- Shortest range
- Low penetrating power (paper)



Beta Decay (b)

- Nucleus ejects beta particle
- When an element undergoes beta decay, one neutron converts to a proton, decreasing the atomic number by one
- Atomic mass doesn't change because the total number of protons & neutrons stays the same
- Equivalent to an electron with a charge of -1
- Stopped by a thin sheet of metal
- Hydrogen undergoes beta decay



Gamma Decay (γ)

- When the nucleus doesn't emit a particle but instead a form of electromagnetic wave known as a gamma-ray
- Equivalent to a high-energy X-ray with a charge of 0
- Travel very far and stopped by a few metres of concrete

Relative abundance:

$$A_r = \frac{(\% \text{ of Isotope 1} \times \text{mass of Isotope 1}) + (\% \text{ of Isotope } n \times \text{mass of Isotope } n)}{100}$$

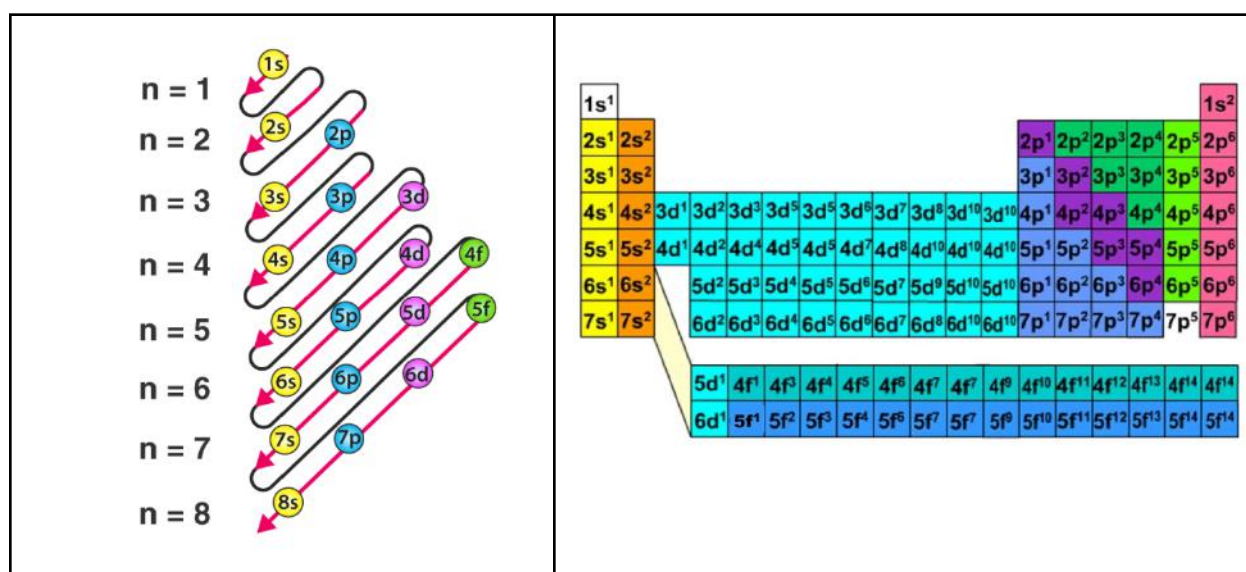
- Percentage of that isotope in its naturally occurring element
 - E.g. 98.89% of Carbon-12 and 1.11% of Carbon-13
 - Atomic mass C = $98.89 \times 12 + 1.11 \times 13 / 100 = 12.01$
- **Ions:** Charged particles
 - If the charge is positive = more protons (losing electrons to form cation)
 - If the charge is negative = more electrons (gaining electrons to form anion)

Electron configuration (SPDF notation)

- Electron shells (named by principal quantum numbers) are divided into subshells (s,p,d,f)

- Subshells are divided into orbitals which are regions around the nucleus where the electron is present)
 - Maximum number is 2 electrons that spin in opposite directions

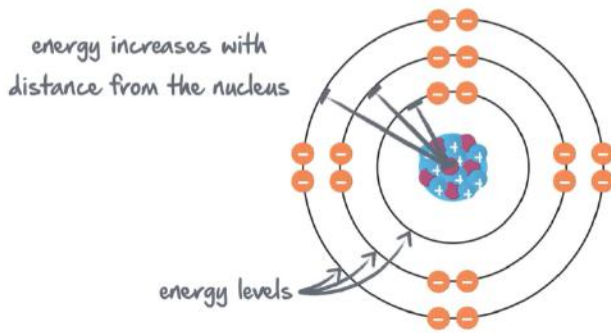
Subshell	No. of orbitals	No. of electrons
S (sphere shape)	1	2
P (dumbbell shape)	3	6
D (clover of four of them)	5	10
F (silly one)	7	14



- Electrons fill from the inner, lower energy shells first
- Electrons don't like being close together because they repel each other
 - Within subshells, electrons first fill up empty orbitals before pairing up
 - According to Aufbau's principle, $4s$ orbitals are filled up before $3d$ orbitals because they require less energy to fill up, and electrons fill the lowest energy orbital first.
 - Copper and Chromium are exceptions to this rule

Bohr model

- The Bohr theory was that electrons move around the nucleus in fixed orbits
- Energy levels further away from the nucleus have a higher level of energy
 - Each electron shell has an energy associated with it



- Electrons can move between the levels meaning the electron has to change its energy to the energy level it moves to
 - **EXCITATION:** When an electron wants to move to a higher energy level it needs to gain energy (excited state = energy level greater than ground state)
 - Electrons absorb photons (packets of light energy/electromagnetic radiation)
 - **DE-EXCITATION:** When an electron wants to move to a lower energy level it needs to lose energy (ground state = energy level it normally occupies)
 - Electrons release photons (wavelengths that determine colour)
- **EXCITED STATE:**
- The amount of energy absorbed/emitted by the electron must equal the difference between the energy levels involved

Spectral evidence

- Line spectra (Patterns of light): a band of colour that is not continuous and is caused by light of particular wavelengths
- 1. **Emission spectrum:** When an atom has electrons in higher levels that can move down to lower levels
 - They emit photons of different energies which correlate to the difference in electron energy levels
 - Some of these photons are visible light
 - Bright-coloured lines against a black background
- 2. **Absorption spectrum:** When white light is shone on an atom, some photons are absorbed to move electrons to higher energy levels
 - Photons absorbed will have energies that match specific wavelengths
 - Continuous spectrum that is missing some colour
- Lines become closer together as the frequency increases
- Higher frequency = smaller wavelength = more energy
 1. **Lynman series:** Ultraviolet range of the electromagnetic spectrum
 2. **Balmer series:** Visible range of the electromagnetic spectrum
 3. **Paschen series:** Infra-red region of the electromagnetic spectrum

Schrodinger

- Treated the electron as a wave instead of a particle
- Schrödinger's model allowed the electron to occupy three-dimensional space

Similarities	Schrodinger	Bohr
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Electrons have a set of discrete (varying) energy levels	Used maths equations for the behaviour of waves to develop a model that describes the likelihood of finding an electron in a particular position at any time (orbitals)	Electrons move around the nucleus in fixed orbits (known orbit and known radius)= FALSE
	-> Schrodinger equation: a region where the electrons are likely to be found (most likely to be where electron cloud is more dense)	Correctly explains the spectra of Hydrogen but it is not accurate for larger atoms

Flame test

- Potassium ions in a flame emit a purple colour. When potassium ions are sprayed into the flame, the electrons of the cations absorb energy from the flame. If the amount of heat absorbed equals the energy difference of two energy levels within a potassium ion, an electron can become excited and move to higher energy levels. Then, when the electrons de-excite, they move to lower energy levels and emit a photon of discrete frequency that is equivalent to the energy difference between the two energy levels. Therefore, discrete wavelengths of light are emitted and if these are in the visible spectrum, like potassium, a colour will be viewed.

ELEMENT	FLAME COLOUR
Lithium	Carminc (dull red)
Sodium	Yellow
Potassium	Light purple (lilac)
Calcium	Brick-red (orange-red)
Strontium	Scarlet (deep red)
Barium	Pale green (apple green)
Copper	Blue-green

Periodicity

- Most elements are solid at room temperature
- Liquids: Mercury and Bromine
- Gases: Noble gases + Hydrogen + Oxygen + Fluorine + Chlorine + Nitrogen
- Elements in the same group have similar reactivity and behave in the same ways because they have the same number of valence electrons
 - The equation to determine the number of electrons in a valence shell is $2n^2$

Effective nuclear charge:

- Measure of attractive forces between valence electrons of an atom and the nucleus

Down a group

- Remains the same down a group because although the number of protons is increasing, the number of inner electrons is also increasing

Across a period

- Increases across a period because the number of protons is increasing, but the number of inner electrons remains the same

Atomic radius

- It is the size of an atom is difficult to measure, so atomic radii are measured using covalent bond lengths for non-metals and metallic bonds for metals
- Positive ions are smaller than the original atom
- Negative ions are larger than the original atom

Down a group

- Atomic radius increases down a group because the effective nuclear charge remains constant, but the number of electron shells increases. This means valence electrons are less strongly attracted to the nucleus and end up further away.

Across a period

- Atomic radius decreases across a period because the number of protons increases, so the effective nuclear charge increases. This means valence electrons are more strongly attracted to the nucleus, so the atomic radius is smaller.

Ionisation energy

- The energy required to remove an electron and make it stable
 - Electron is removed from the highest energy level because they require less energy to escape the pull of the nucleus because they have more energy
- In graphs of ionisation energy, big jumps occur when shells are removed

First Ionisation energy

- The energy required to remove one electron to form an ion of +1 charge (gas + moles)

Second Ionisation energy

- The energy required to remove one electron to form an ion of +2 charge (gas + moles)
 - The energy required to remove one electron after the first ionisation energy
- Larger than FIE because when valence electrons are removed, a shell is lost (less electron shielding), meaning attraction between electrons and nucleus is larger and requires more energy to remove

Successive ionisation energy

- The charge of the electron increases by 1 and 1 electron is removed

Down a group

- First Ionisation energy decreases down a group because the atomic radius increases and more shells are added, meaning the valence electrons are further away from the nucleus. This increases the effect of electron shielding and decreases the attraction between valence electrons and the nucleus, making it easier for an electron to be removed.

Across a period

- First Ionisation energy increases across a period because the number of protons increases, increasing the effective nuclear charge. This means attraction between the nucleus and valence electrons increases, making it harder for an electron to be removed. Extra electrons added are in the same energy level and therefore don't experience much electron shielding.

Electronegativity

- The ability of atoms of an element to attract a bonding pair of electrons
- Non-metals are more electronegative because they need to gain electrons to become stable
- Noble gases don't have electronegativity because they have a full valence shell and are stable, meaning they don't need to attract electrons.
- The higher the electronegativity, the more reactive for non-metals

Down a group

- Electronegativity decreases down a group because the atomic radius increases and more shells are added, meaning the valence electrons are further away from the nucleus. This increases the effect of electron shielding and decreases the attraction between valence electrons and the nucleus, meaning bonding electrons are less strongly attracted to the nucleus.

Across a period

- Electronegativity increases across a period because the number of protons increases, increasing the effective nuclear charge. This means attraction between the nucleus and valence electrons increases, and bonding electrons are more strongly attracted to the nucleus. Extra electrons added are in the same energy level and therefore don't experience much electron shielding.

Reactivity with water

- Group 1 (Alkali metals) react vigorously with water to form hydrogen gas and a solution
- Group 2 (Alkaline earth metals) react with water but less vigorously than group 1 to form hydrogen gas and an alkaline solution
 - Beryllium and Magnesium need steam to react vigorously
- Groups 3-6 hardly react with water
- Group 7 (Halogens) react with water to form acids because they are very electronegative
- Group 8 does not react with water

Reactivity of metals with water

Down a group

- Increases down a group because First Ionisation energy decreases

Across a period

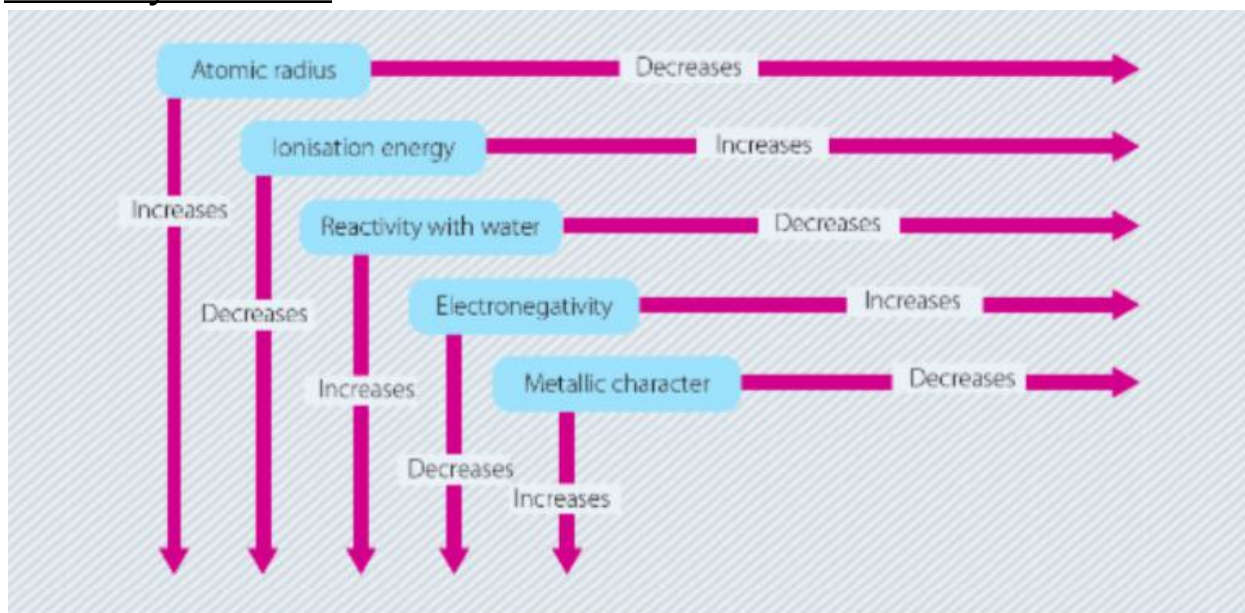
- Decreases across a period because First Ionisation energy increases

Reactivity of group 7 with water

Down a group

- Decreases down a group because electronegativity decreases

Summary of trends



Chemical bonding

Intramolecular bonding

- Refers to bonding between atoms (determines chemical properties)

Ionic bonding

- Ionic bonding is a type of chemical bonding that involves the transfer of electrons from one atom to another
 - Strong electrostatic attraction between positive and negative ions which holds the ions together in ionic bonding
- An infinite array of positive and negative ions (no discrete molecules)
- Overall neutral charge (total number of positive charges = negative charges)
- Naming: Cation, anion ending in "ide"
- Empirical formula (simplest ratios)
- Lattice structures

Properties of ionic substances

- **High melting/boiling points** because of strong electrostatic attraction between ions that is difficult to overcome and requires a lot of energy
- **Hard** because of the strong electrostatic attraction between ions
- **Brittle** because when the orderly array of ions is disturbed by applying a strong force, ions of the same charge come close together and repel each other, causing the crystal to shatter.
- **Poor electrical conductors as solids** because they are fixed in an array and electrons are unable to carry the charge
 - When **molten**, the orderly arrangement of ions is broken up and the ions can move freely and carry a charge, and therefore can conduct electricity.
 - When **dissolved** in water, the crystals are completely broken up and the solutions consist of individual ions moving randomly that can migrate towards oppositely charged electrodes, and therefore conduct electricity.
 - "Mobile charge carriers"
- **GENERALLY soluble in water** because water is polar where oxygen is partially negative and hydrogen is partially positive. The ions are strongly attracted to the oppositely charged sides

of the polar water molecules that pull the ions away from the lattice and cause them to dissolve.

- "Hydration spheres"
- Ion-dipole forces

Polyatomic ions

Ion	Formula	Valency
Ammonium	NH_4^+	1
Ammonia	NH_3	NEUTRAL
Hydroxide	OH^-	1
Sulphate	SO_4^{2-}	2
Nitrate	NO_3^-	1
Carbonate	CO_3^{2-}	2
Sulfuric acid	H_2SO_4	6
Phosphate	PO_4^{3-}	3

Covalent bonding

- Covalent bonds are formed between pairs of atoms by the atoms sharing electrons
 - Between semi-metal/non-metal and nonmetals
- Naming: 1) Element with lower electronegativity 2) Element and "ide"
 - Prefixes for the number of atoms for elements (no 'mono' for the first element)
 - Mono, di, tri, tetra, penta, hexa, hepta, octa, nona

Properties of covalent molecular substances

- Molecules are held together by strong covalent bonds and attracted to each other by weak intermolecular forces
 - **Low melting/boiling point** because only the weak intermolecular forces need to be overcome (not the covalent bonds)
 - **Soft** because of weak intermolecular forces that are easy to break
 - **Don't conduct electricity** because they are neutral and there are no charged particles
 - **Varied solubility** (polar = soluble, non-polar = insoluble)

Properties of covalent network/lattice

- Strong covalent bonds that extend through the entire lattice (E.g. diamond)
 - **High melting/boiling points** because of strong covalent bonds that are difficult to overcome
 - **Brittle** and **hard** because the strong covalent bonds are stiff (at specific angles).

- For diamond, they are clear and transparent because of the orderly arrangement of electrons.
- **Poor electrical conductors** because electrons can't move and are in covalent bonds
- Insoluble in water because the covalent bonds are stronger than the bonds that form between water and the network, meaning it won't dissolve
- Graphite is soft and can conduct electricity because it is made of sheets of carbon atoms, with weak intermolecular bonds between each sheet that are easy to break. Spaces between the sheets are filled with delocalised electrons that conduct electricity along its plane.

Electronegativity

- Some elements can form ionic and covalent compounds
- If the difference in electronegativity is greater than 1.5, the compound is ionic
- If the difference in electronegativity is less than 1.5, the compound is covalent

Electronegativity difference	Distribution of bonding electrons	Type of bond	Example
zero	electrons shared equally	non-polar covalent	H ₂ , Cl ₂ , CS ₂
less than 1.7	electrons attracted to the more electronegative atom	polar covalent	NH ₃ , H ₂ O, HCl
greater than 1.7	electrons transferred to the more electronegative atom	ionic	NaCl, CaF ₂

Metallic bonding

- Solids at room temperature (except mercury)
- A 3D lattice of positive ions (cations) immersed in a 'sea' of delocalised electrons
- The valence electrons break away from their atoms, leaving behind positive ions
 - Free electrons are delocalised and move randomly throughout the lattice
- Strong electrostatic attraction between delocalised electrons and the ions of the lattice holds the metal together

Properties of metals

- **Malleable** and ductile arise because when the orderly array of positive ions is sheared, the mobile electrons can adjust to the new arrangement of positive ions and provide the bonding that holds the whole lattice together.
- **Good electrical conductors** because delocalised electrons can move freely through lattice
- **Hard** because of the strong electrostatic attraction between positive ions and electrons
- Most have relatively **high melting points** because metallic bonds are strong
- **Insoluble in water** because the metallic bond is stronger than the bonds that form between water and the metal, meaning it won't dissolve
- **High density** because they have many atoms packed very closely into a small space

Intermolecular forces and allotropy

Polarity

- Refers to the uneven sharing of electrons because of different electronegativities in the elements of the compound
 - The electrons will be greater attracted to elements with greater electronegativity

- E.g. HCl (Cl has a partial negative charge + H has a partial positive charge)
- **Dipoles:** A pair of equal and opposite charges separated in space (represented by a delta)
 - Arrows point from the positive end to the negative end of a dipole



Dipole Moment has a **Magnitude** and a **Direction**

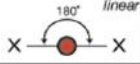
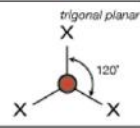
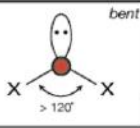
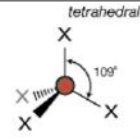
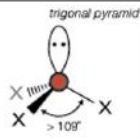
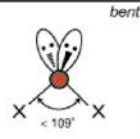
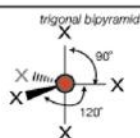
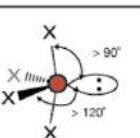
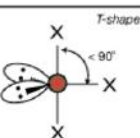
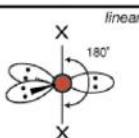
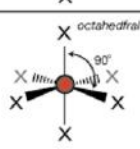
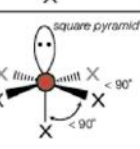
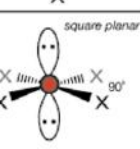
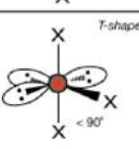
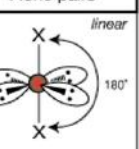
- **Polar molecules:** compounds with a net dipole (slightly negative and slightly positive ends)
 - Diatomic polar molecules:
 - Polyatomic polar molecules: Sometimes a molecule may have a 'dipole' moment but does not have a net overall dipole
 - Despite having polar bonds, the symmetry of the molecules means that the dipoles cancel out and the molecule is nonpolar
 - Substances with similar polarities tend to be soluble in one another
 - Non-polar: Soluble in oils and lipids
 - Polar: Soluble in water and alcohols

Shapes of molecules

Valence Shell Electron-Pair Repulsion theory (VSEPR)

- Pairs of valence electrons arrange themselves to be as far away as possible from other electron pairs because electrons repel each other and want to get the most stable electron configuration
 - Lone pairs repel each other to a greater extent
- **Double bonds:** Some molecules form two covalent bonds between atoms.
 - Represented by two covalent bond lines in a Lewis structure diagram

TOTAL NUMBER OF ELECTRON PAIRS	ARRANGEMENT OF ELECTRON PAIRS	BONDING PAIRS	LONE PAIRS	SHAPE OF MOLECULE	EXAMPLES
2	Linear	2	0	Linear	BeCl ₂
3	Trigonal planar	3	0	Trigonal	BCl ₃
4	Tetrahedral	4	0	Tetrahedral	CH ₄ , SiF ₄
		3	1	Pyramidal	NH ₃ , PCl ₃
		2	2	Bent	H ₂ O, SCl ₂
5	Trigonal bipyramidal	5	0	Trigonal bipyramidal	PCl ₅
6	Octahedral	6	0	Octahedral	SF ₆

0 lone pairs		1 lone pair		2 lone pairs		3 lone pairs		4 lone pairs	
Number of atoms around the central atom (steric number)	2								
	3								
	4								
	5								
	6								

Identifying the shape of molecules

1. Work out which is the central atom (atom there is one of)
2. Work out the number of valence electrons of the central atom
3. Work out the number of electrons involved in bonding with other atoms by using its molecular formula
4. Work out the number of lone electron pairs of the central atom by dividing the number of left-over electrons by two
5. Work out the shape

Intermolecular forces

- The polarity of molecules affects the strength of intermolecular forces
- **Intermolecular forces:** forces between molecules or ions (determine physical properties)
 - **Weaker** than **intramolecular** forces because it takes more energy to overcome covalent bonds and break apart molecules than to overcome intermolecular forces
 - When a substance changes state, the molecules remain intact
 - Intermolecular forces dictate changes in state, not intramolecular forces

Dipole-dipole forces

- Only exists between polar molecules because it's the attraction between the positive end of a polar molecule and the negative end of another polar molecule
 - For molecules of approximately equal mass and size, the strength of dipole-dipole forces increases with polarity
- Only effective when polar molecules are very close (when the distance between polar molecules increases, the strength of the bond decreases)
- Molecules with large dipole moments have higher melting/boiling points because bonds are harder to break

Hydrogen bonding

- Special type of dipole-dipole bonding between a hydrogen covalently bonded with Fluorine, Oxygen or Nitrogen and a lone pair of electrons in Fluorine, Oxygen or Nitrogen atoms
 - Create large dipoles because they are very electronegative and have a small atomic radius
- Around 10 times stronger than dipole-dipole forces

London dispersion forces

- All molecules have some degree of dispersion forces (polar and nonpolar molecules)
- Weakest intermolecular force

Temporary dipole/instantaneous dipole

- Electrons temporarily shift to one end of a molecule and create a temporary dipole
 - For a brief amount of time, there is an unequal distribution of electrons, and since they are charged, the side with more electrons is partially negative

Induced dipole

- The area of partial negative charge will repel other electrons in adjacent molecules creating a new temporary dipole

Dispersion force: Areas of opposite charge from temporary dipoles in adjacent molecules will be attracted to each other (when the electrons move back, the dipoles disappear)

- Dispersion forces are usually weak unless they are really big
- An increase in molecular weight = larger molecules = more number of electrons = easier it is to produce a temporary dipole = the greater the dispersion forces
- Linear molecules form stronger dispersion forces because they have a larger contact area for interactions between molecules to occur and therefore stronger dispersion forces

Allotropy

- Existence of two or more physically distinct forms of one element in the same state
 - Different physical properties

Carbon allotropes

ALLOTROPE	MELTING POINT (°C)	CONDUCTS ELECTRICITY?	SOLUBLE IN CYCLOHEXANE?
Graphite	3974	Yes	No
Diamond	3550	No	No
Buckminsterfullerene (C ₆₀)	550	?	Yes

Diamond

- Each carbon bonds with four other carbon atoms (tetrahedral structure)
- Electrons are strongly fixed in structure and are immobile (insulator)

Graphene

- Each carbon bonds with three other carbon atoms
 - Electrons aren't fixed and are mobile (good conductors)
 - Graphite: sheets of graphene bound by weak intermolecular forces (soft)
- Used for nanoelectronics and battery technology

Fullerenes

- (C₆₀) spherical with hexagonal patches surrounded by 3 hexagonal and pentagonal patches
- Interstellar space dust

MODULE 2- Introduction to quantitative chemistry

Chemical Reactions and Stoichiometry

Inquiry question: What happens in chemical reactions?

- conduct practical investigations to observe and measure the quantitative relationships of chemical reactions, including but not limited to:
 - masses of solids and/or liquids in chemical reactions
 - volumes of gases in chemical reactions

- Relate stoichiometry to the law of conservation of mass in chemical reactions by investigating:
 - balancing chemical equations
 - solving problems regarding mass changes in chemical reactions

Mole Concept

Inquiry question: How are measurements made in chemistry?

- Conduct a practical investigation to demonstrate and calculate the molar mass (mass of one mole) of:
 - an element
 - a compound

- Conduct an investigation to determine that chemicals react in simple whole number ratios by moles

- Explore the concept of the mole and relate this to Avogadro's constant to describe, calculate and manipulate masses, chemical amounts and numbers of particles in:
 - moles of elements and compounds $n = m/MM$ (n = chemical amount in moles, m = mass in grams, MM = molar mass in g mol^{-1})
 - percentage composition calculations and empirical formulae
 - limiting reagent reactions

Concentration and Molarity

Inquiry question: How are chemicals in solutions measured?

- Conduct practical investigations to determine the concentrations of solutions and investigate the different ways in which concentrations are measured

- Manipulate variables & solve problems to calculate concentration, mass or volume using:
 - $c = n/V$ (molarity formula)
 - dilutions (number of moles before dilution = number of moles of sample after dilution)
- Conduct an investigation to make a standard solution and perform a dilution

Gas Laws

Inquiry question: How does the Ideal Gas Law relate to all other Gas Laws?

- Conduct investigations and solve problems to determine the relationship between the Ideal Gas Law and:
 - Gay-Lussac's Law (temperature)
 - Boyle's Law
 - Charles' Law
 - Avogadro's Law

Chemical reactions, stoichiometry and the mole

Law of conservation of mass

- States that matter can't be created nor destroyed, but changed from one form to another
- This means that in any chemical reaction mass is conserved
 - total mass of products = total mass of reactants
 - the number of atoms of each element is conserved
- A balanced chemical equation must have the same number of atoms of each element in the reactants and products

Relative atomic mass - Relative atomic mass is the average mass of the atoms present in the naturally occurring element relative to the mass of an atom of the carbon-12 isotope	Calculate the relative atomic mass of naturally occurring chlorine-35 of relative abundance 75% and chlorine-37 of relative abundance 25% ANSWER $\text{Average mass of 100 atoms} = \frac{75 \times 35 + 25 \times 37}{100}$ $= 35.5$ Therefore relative atomic mass of chlorine is 35.5.
Relative molecular mass - The relative molecular mass of a covalent molecular compound is the mass of a molecule of the compound relative to the mass of an atom of the carbon-12 isotope	Relative molecular mass of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ ANSWER $M = 12 \times A_{\text{C}} + 22 \times A_{\text{H}} + 11 \times A_{\text{O}}$ $= 12 \times 12.01 + 22 \times 1.008 + 11 \times 16.00$ $= 342.3$
Relative formula mass - The relative formula mass of a compound is the sum of the relative atomic masses of the atomic species as given in the stated formula of the compound. - For ionic compounds that don't have discrete molecules but exists in ratios	Relative formula mass of calcium phosphate, $\text{Ca}_3(\text{PO}_4)_2$ ANSWER $M = 3 \times A_{\text{Ca}} + 2 \times (A_{\text{P}} + 4 \times A_{\text{O}})$ $= 3 \times 40.08 + 2 \times (30.97 + 4 \times 16.00)$ $= 310.2$

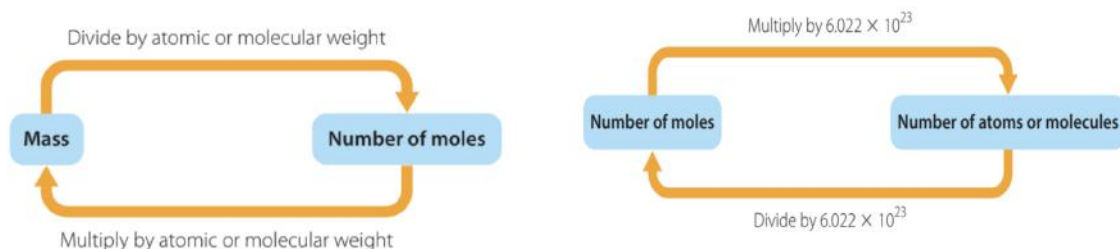
Particle Theory

- 1) All matter is made out of particles

- 2) Particles have space between them
- 3) The particles are always moving unless theoretically below absolute zero (0K)
- 4) Particles move faster and get further apart when heated
- 5) Particles are attracted to each other
- 6) All particles of the same substance are identical

The mole

- A **mole** of a substance is the quantity that contains as many particles as there are atoms in exactly 12g of the carbon-12 isotope
- A mole of gaseous elements can refer to their atoms or molecules (diatomic)
- $n = N/N_A$ (find molecules)
 - To find atoms multiply by how many atoms there are
- **Avogadro constant:** the number of atoms in exactly 12g of the carbon-12 isotope
- **Molar mass:** the mass of a mole of the substance (mass/mole or gmol^{-1} or g/mol)



Percentage composition

- Number of grams in one mole of a compound

The general method for calculating percentage composition is:

$$\%A \text{ in a compound} = \frac{\text{mass of A in one mole of the compound}}{\text{mass of one mole of the compound}} \times 100$$

In a compound of formula $A_wB_yC_z$:

$$\%A = \frac{w \times (\text{relative atomic mass of A}) \times 100}{\text{relative molecular or formula mass of } A_wB_yC_z}$$

Formulae from experimental data

Empirical formula

- Simplest whole-number ratio in which the atoms are present in a compound
1. Masses of all elements present in a given sample of the compound
 2. Convert masses to moles
 3. Divide by the smallest number of moles to get a simple ratio
 4. Round off the numbers to get whole numbers and write the empirical formula

Magnesium oxide experiment:

- A lid is used because the burning of magnesium produces so much heat that it can break the crucible (limits the heat)

- The crucible and lid were heated and reweighed to get rid of impurities that would affect the weight of the crucible
- Magnesium ribbon was sanded thoroughly to increase SA/V ratio (allows for a more accurate formula) because a SA/V ratio affects oxidation/reaction of magnesium

Experimental value may differ from theoretical value:

- 1) Systematic errors
 - Magnesium is not pure
 - Not all of the magnesium weighed reacted
 - The product was not magnesium oxide only
 - Loss of magnesium or of magnesium oxide occurred
 - Change in the mass of the crucible occurred during handling/heating
 - Insufficient heat energy because magnesium ribbon was not rolled properly or placed on the bottom of the crucible

Molecular formula

- Include the actual number of atoms of elements present in a compound
 1. Molecular mass = $n \times \text{empirical mass}$
 2. Find n and multiply it with the coefficient of every element in the compound
 - a. n is the HCF
- Ionic compounds have the same empirical formula and molecular formula

The law of definite composition

- Compounds always contain the same elements in constant proportion by mass
- E.g. NaCl is always composed of 39.3% sodium and 60.7% chlorine

Limiting reagent

- Reactant that is used up first in the reaction.
- Hence, it limits the amount of product formed.
 - Amount of product determined by limiting reagent
 - Use moles of limiting reagent in the stoichiometric ratio

Solutions- concentration and molarity

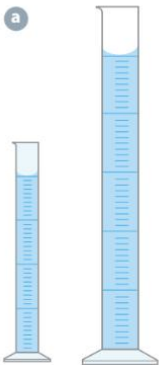
- **Solution:** homogeneous mixture of two or more substances
 - **Solvent:** Substance that does the dissolving
 - **Solute:** Substance that dissolves
- A **diluted solution** contains a relatively low concentration of the solute
- A **concentrated solution** contains a relatively high concentration of solute

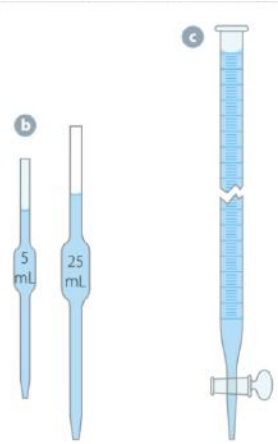
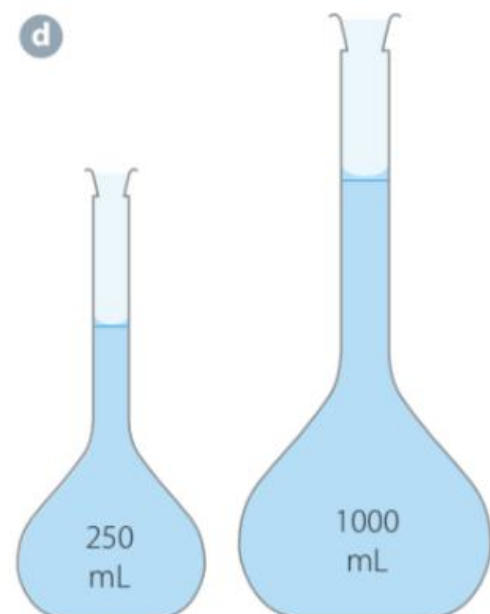
Measures of concentration

- **mass** of solute per 100mL or 1 L of **solvent**
- **mass** of solute per 100mL or 1 L of **solution**
- **volume** of solute per 100mL or 1 L of **solvent**
- **volume** of solute per 100mL or 1 L of **solution**

%(w/v) or 'per cent, weight, volume' – mass of solute per 100mL of solution	1 A 0.90%(w/v) solution contains 0.90 g NaCl in 100 mL of solution. $\text{Mass of NaCl required in 250 mL} = \frac{250}{100} \times 0.90$ $= 2.3 \text{ g}$
%(v/v) or 'per cent, volume, volume' – volume of solute per 100mL of solution	2 In 100 mL of solution there are 14 mL of alcohol. $\text{Volume of alcohol in 750 mL} = \frac{750}{100} \times 14$ $= 105 \text{ mL}$ $= 1.1 \times 10^2 \text{ mL}$
%(w/w) or 'per cent by weight' – mass of solute per 100g solution - ppm or 'parts per million' – grams of solid or liquid solute per million grams of solution, while for gases it means molecules per million molecules. - If % is used on its own, it generally means %(w/w)	3 500 ppm means 500 g mercury(II) ion per million grams of solution; that is, per 10^6 g solution. %(w/w) means grams per 100 g solution. $\text{Mass of mercury(II) ion in 100 g} = \frac{100}{10^6} \times 500$ $= 0.050 \text{ g}$ Therefore the solution is 0.050%(w/w).

Measuring volumes of a solution

Measuring cylinders - Approximate amounts are required - Volumes from measuring cylinders are accurate only to about $\pm 5\%$ - Common sizes are 10, 100, 250 and 500mL	
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Pipettes and burettes - Deliver accurate volumes to about ± 0.2 to $\pm 0.5\%$ Pipettes (b) - Deliver fixed volumes - Common pipette sizes are 5, 10 and 25mL, (1, 2 and 50mL sizes are available) Burettes (c) - Deliver variable volumes, generally from 0 to 50mL - One drop at a time (valve used to control quantity) - Allows you to add very small quantities at a time (add, stir, observe any reaction).	
Volumetric flasks - Contain fixed volumes accurate to about ± 0.2 to $\pm 0.5\%$ when filled to the mark engraved on their necks - Common sizes are 100, 250, 500mL and 1L, although 50mL and 2L sizes are also available - Fill volumetric glassware until the bottom of the meniscus sits accurately on the graduation mark - To make an aqueous solution of accurately known concentration (better than $\pm 1\%$) 1. Measure out the required amount of solute (using a sensitive balance for mass or pipette for volume) 2. Transfer it quantitatively to a volumetric flask 3. Dissolve it in some water 4. Fill the flask to the graduation mark. ➤ Concentration in mass or volume per unit volume of solution	

Molarity

Standard solutions

- Solutions in which the concentration is accurately known - $C = n/V$	$\text{Molarity} = \frac{\text{number of moles of solute}}{\text{volume of solution in litres}}$
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Precise amount of solute

- **Primary standards: (High PANTS)**

- Type of solute used in standard solutions
- Truly representative of the number of moles of substances



Primary Standards

Property	Why?
High Molecular Mass	Minimises impact that errors in weighing have on calculation of moles present
Purity	If not pure, the mass of a sample would be too high compared to how many moles of solute there are
Affordability and availability	To be able to source and purchase the primary standard
Non-toxicity	Primary standard should be as safe as possible to handle
Stability	Ensures that Primary Standards don't react easily

1. Wash watch glass and beaker with deionised water and dry completely
2. Weigh watch glass and beaker
3. Weigh solute on electronic balance
4. Transfer solute into a beaker to stir and add a little bit of water to dissolve before transferring in the volumetric flask
5. Pour solute into a volumetric flask using a clean funnel
6. Rinse any remaining solute particles from beaker/funnel into volumetric flask using deionised water

Precise volume of solution

1. Dissolve solute by filling the volumetric flask with deionised water. Add a stopper and swirl the volumetric flask
2. Add deionised water up to the meniscus of the volumetric flask. Use a wash bottle, then a pipette for final drops.
 - a. **Rinse pipette with solution to ensure that any water or solution from previous use is completely removed**
3. Add a stopper and shake flask to ensure solution is homogeneous

Dilution

(concentration of diluted solution) × final volume of diluted solution = (original concentration) × volume of original solution	$c_2V_2 = c_1V_1$
$V_2 = (c_1V_1)/c_2$	$c_2 = (c_1V_1)/V_2 + V_1$

Gas Laws

The Ideal Gas

- A theoretical gas is composed of many randomly moving particles that are not subject to interparticle interactions
- **They ideally have no volume**
- Bigger particles are less ideal
- $\text{He}_2(\text{g})$ is the closest gas to the ideal

Real Gas vs Ideal Gas

REAL GAS

- Particles have volume
- Energy lost in collisions
- Intermolecular forces



IDEAL GAS

- Particles have no volume
- Collisions are elastic
- No interactions between particles



Real gases behave like ideal gases:

- **At high temperatures**
- **At low pressures**

sciencenotes.org

Pressure

- Pressure is the force exerted on the unit area of a surface by the particles of a gas
 - $P = \text{pressure (kPa)}$
 - $F = \text{force (N)}$
 - $A = \text{area (m}^2\text{)}$
- **Dalton's Theory:** The pressure exerted by a mixture of gas is equal to the sum of the pressures that would be exerted by each gas alone in the same volume.

$$P_{\text{total}} = P_1 + P_2 + P_3 \dots$$

Units of pressure

$$\text{Kilo Pascals} = \text{Atmospheric Pressure} \times 101.325$$

Temperature

$$\text{Kelvin} = \text{Celsius} + 273.15$$

- **Absolute temperature scale:** All physically accessible temperatures have positive values. 0K is the absolute zero of temperature

Universal Gas Constant

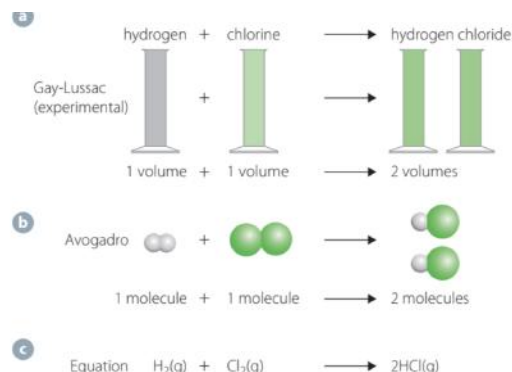
- $R = 8.314$, if pressure = kPa and volume = L ← (8pi)
- $R = 0.08206$, if pressure = atm and volume = L

Avogadro's Law: Volume and moles

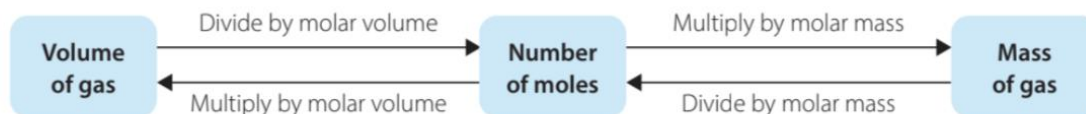
When measured at the **same temperature** and **pressure**, equal volumes of different gases contain the same number of molecules

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

- V_1 = initial volume (L)
- V_2 = final volume (L)
- n_1 = initial moles (mol)
- n_2 = final moles (mol)



- **Gay-Lussac's law of combining volumes:** When measured at constant temperature and pressure, the volumes of gases taking part in a chemical reaction show simple whole-number ratios to one another.
- **Molar volume of gas:** Because a mole is a fixed number of molecules, a mole of any gas has the same volume as a mole of any other gas (at the same temperature and pressure).
 - At 0°C and exactly 100kPa, the molar volume of all gases is 22.71L
 - At 25°C and exactly 100kPa, the molar volume of all gases is 24.79L

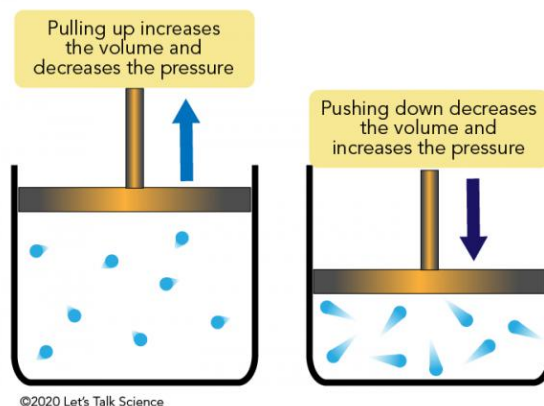


Boyle's Law: Volume and Pressure

For a given quantity of gas at a constant temperature, the product of its volume, V, and its pressure, P, is constant. Volume is inversely proportionate to the pressure, $PV=k$

$$P_1 V_1 = P_2 V_2$$

- P_1 = initial pressure (kPa)
- V_1 = initial volume (L)
- P_2 = final pressure (kPa)
- V_2 = final volume (L)



$$\rightarrow P \propto \frac{1}{V}$$

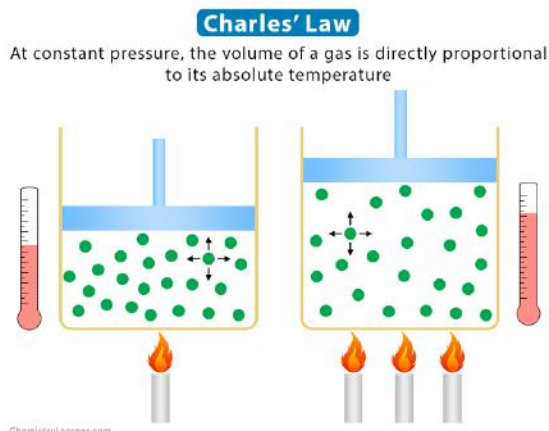
Charles' Law: Volume and Temperature

For a fixed quantity of gas at constant pressure, volume increases linearly with temperature. $V \propto T$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

- V_1 = initial volume (L)
- T_1 = initial temperature (K)
- V_2 = final volume (L)
- T_2 = final temperature (K)

$$\rightarrow V \propto T$$



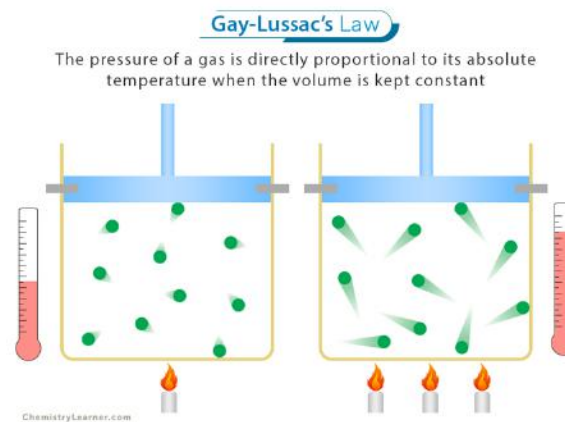
Gay-Lussac's Law: Temperature and Pressure

For a fixed sample of gas at constant volume, pressure increases linearly with temperature. $P \propto T$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

- P_1 = initial pressure (kPa)
- T_1 = initial temperature (K)
- P_2 = final pressure (kPa)
- T_2 = final temperature (K)

$$\rightarrow P \propto T$$

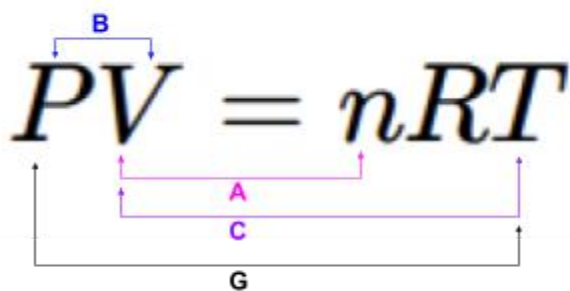


Combined gas law

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Ideal gas law

$$PV = nRT$$



Gas Laws

Can - Charles's
 These - Temp.
 Guys - Gay-Lussacs
 Possibly - Pressure
 Be - Boyle's
 Victorious - Volume

Module 3- Reactive chemistry

Chemical Reactions

Inquiry question: What are the products of a chemical reaction?

- Investigate a variety of reactions to identify possible indicators of a chemical change
- Use modelling to demonstrate
 - the rearrangement of atoms to form new substances
 - the conservation of atoms in a chemical reaction

- Conduct investigations to predict and identify the products of a range of reactions, for example:
 - synthesis
 - decomposition
 - combustion
 - precipitation
 - acid/base reactions
 - acid/carbonate reactions

- Investigate the chemical processes that occur when Aboriginal and Torres Strait Islander Peoples detoxify poisonous food items
- Construct balanced equations to represent chemical reactions

Predicting Reactions of Metals

Inquiry question: How is the reactivity of various metals predicted?

- Conduct practical investigations to compare the reactivity of a variety of metals in:
 - water
 - dilute acid
 - oxygen
 - other metal ions in solution

- Construct a metal activity series using the data obtained from practical investigations and compare this series with that obtained from standard secondary-sourced information

- Analyse patterns in metal activity on the periodic table and explain why they correlate with, for example:
 - ionisation energy
 - atomic radius
 - electronegativity

- Apply the definitions of oxidation and reduction in terms of electron transfer and oxidation numbers to a range of reduction and oxidation (redox) reactions
- Conduct investigations to measure and compare the reduction potential of galvanic half-cells
- Construct relevant half-equations and balanced overall equations to represent a range of redox reactions
- Predict the reaction of metals in solutions using the table of standard reduction potentials
- Predict the spontaneity of redox reactions using the value of cell potentials

Rates of Reactions

Inquiry question: Are there patterns in the properties of elements?

- Conduct a practical investigation, using appropriate tools (including digital technologies), to collect data, analyse and report on how the rate of a chemical reaction can be affected by a range of factors, including but not limited to:
 - temperature
 - surface area of reactant(s)
 - concentration of reactant(s)
 - catalysts

- Investigate the role of activation energy, collisions and molecular orientation in collision theory
- Explain a change in reaction rate using collision theory

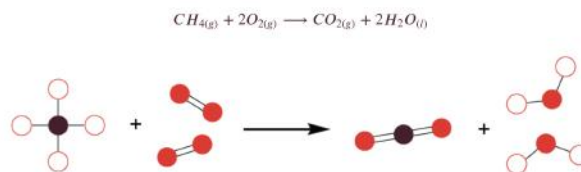
Types of chemical reactions

Indicators of a chemical reaction

- A gas is evolved
- A precipitate is formed
- A change in colour occurs
- A significant change in temperature
- A solid disappears
- An odour is produced

Modelling

- The ball and stick model is what you'll be using to represent chemical reactions in practicals. 'Balls' represent the atoms while the bonds are depicted as 'sticks'



Advantages	Limitations
Demonstrates the rearrangement of atoms to form new substances as old bonds are broken and new bonds are formed to produce new products	Doesn't show unshared electron pairs around any of the atoms
Demonstrates the conservation of atoms in a chemical reaction (law of conservation of mass)	Strength of the bond
	Size of the atoms or movement of electrons

Synthesis reactions

- Two or more substances combine to form a new substance
 - E.g. $2\text{Cu(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{CuO(s)}$
- **Direct combination reactions:** two or more pure elements
 - Non-metals react with metals to form ionic compounds
 - Metals do not react with each other chemically (make alloys $\rightarrow$ mixture of metals)
 - Non-metals react with non-metals to form covalent compounds

Decomposition reactions

- Compounds that are decomposed into two or more simpler substances
- Generally endothermic reactions
- 1. **Thermal decomposition (decomposition by heating)**
 - E.g. $\text{CaCO}_3\text{(s)} \rightarrow \text{CaO(s)} + \text{O}_2\text{(g)}$
 - Compounds that can easily be decomposed by heating include
 - Most **hydroxides** (to form oxides and water)
 - Metal hydroxide $\rightarrow$ metal oxide + H_2O
 - Most **carbonates** (to form oxides and carbon dioxide)
 - Metal carbonate $\rightarrow$ metal oxide + CO_2
 - Many **nitrates** (to form oxide, nitrogen dioxide and oxygen)
 - Metal nitrate $\rightarrow$ metal oxide + NO_2 + O_2
- 2. **Electrolysis**
 - An electric current passed through a liquid/solution $\rightarrow$ a chemical reaction)
 - E.g. Electrolysis of water: $2\text{H}_2\text{O(l)} \rightarrow 2\text{H}_2\text{(g)} + \text{O}_2\text{(g)}$
 - E.g. Electrolysis of aqueous copper bromide solution: $\text{CuBr}_2\text{(aq)} \rightarrow \text{Cu(s)} + \text{Br}_2\text{(g)}$
 - Complex reactions so it is difficult to predict products
- 3. **Decomposition by light (basis of film photography)**
 - Silver chloride is a white solid but when exposed to sunlight it changes to purple, then black
After several hours of exposure, the sample has a smaller mass than the starting material
 - Mass decreases because Cl_2 gas dissipates into the environment
 - $2\text{AgCl(s)} \rightarrow 2\text{Ag(s)} + \text{Cl}_2\text{(g)}$

Combustion reactions

- Substances burn in oxygen (or in air) at high temperatures to release energy (heat/light)
- E.g. $2\text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{MgO(s)}$

- When **hydrocarbons** and **cellulose** (contains hydrogen and carbon) undergo **complete combustion**, the products are **carbon dioxide** and **water**
- **Incomplete combustion** (supply of oxygen is limited) produces carbon monoxide and/or carbon soot

Precipitation reactions

- A **precipitate** (solid) forms when two solutions are mixed
- The driving force behind a precipitation reaction is the formation of an insoluble substance
 - When solutions of two ionic substances are mixed, precipitation occurs if one type of positive ion present can combine with one type of negative ion present to form an insoluble substance
- Precipitate *falls* out of solution

Chemical equations

1. **Complete formula equation/neutral species equation:** full balanced chemical equation
2. **Complete/full ionic equation:** all the ions that are present in the reaction mixture
3. **Net ionic equation:** the ions that actually undergo a reaction/participating
 - a. **Spectator ions:** not involved in the reaction
 - b. Shows the exact ion species involved in the chemical reaction, i.e what actually reacts

Solubility

Soluble compounds	Insoluble compounds
Ammonium and Group 1	Carbonates (except group 1 and NH_4^+)
Nitrates	Hydroxides (except group 1, NH_4^+ , Ba^{2+} , Sr^{2+} , Ca^{2+})
Chlorides (except Ag^+ and Pb^{2+})	Oxides (except group 1, NH_4^+ , Ba^{2+} , Sr^{2+} , Ca^{2+})
Sulphates (except Ag^+ , Pb^{2+} , Ba^{2+} , Sr^{2+} , Ca^{2+})	Sulphides (except Na^+ , K^+ , NH_4^+ , Mg^{2+} , Ca^{2+} , Ba^{2+})

- Sometimes when only a **small amount** of a precipitate is formed, it does not settle to the bottom of the test tube as an obvious solid, but stays in **suspension** and makes the mixture look **cloudy** or turbid.
- **Data sheet solubility constants** contain compounds that are insoluble

Acid-base reactions

Acids	Bases
- A substance in solution that produces hydrogen ions (H^+) - In aqueous solution hydrogen ions exist as H_3O^+, an H^+ attached to a water molecule (hydronium ions)	- A substance that contains either the hydroxide ion (OH^-) or the oxide ion (O^{2-}) or produces the hydroxide ion in a solution - Soluble in water = alkalis (sodium and potassium hydroxides)
Sour taste	Bitter taste
Conduct electricity when in solution	Conduct electricity when in solution

Change blue litmus to red	Change red litmus to blue
Sting/burn skin	Soapy feel
Common acids are hydrochloric acid (HCl) , sulfuric acid (H₂SO₄) , nitric acid (HNO₃) , carbonic acid (H₂CO₃) , acetic acid (CH₃COOH) and phosphoric acid (H₃PO₄)	Common bases are sodium hydroxide (NaOH), barium hydroxide (Ba(OH) ₂) magnesium oxide (MgO) iron(III) oxide (Fe ₂ O ₃) copper hydroxide (Cu(OH) ₂) and ammonia (NH ₃)

Neutralisation reaction

- acid + base → salt + water
 - E.g. $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$
- **Bases** - carbonates, hydroxides, oxides
- **Oxyacids**: have oxygen attached to elements like sulphur, nitrogen, phosphorus or carbon
 - When the name ends in '-ic' = the salt ends in '-ate'
 - Anions formed from **oxyacids** are called **oxyanions**

Acid-carbonate

- carbonate + acid → carbon dioxide + water + salt
 - E.g. $\text{CuCO}_3\text{(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{CuSO}_4\text{(aq)} + \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$

Testing for carbon dioxide

- Bubble the gas through clear limewater (calcium hydroxide solution)
- Carbon dioxide turns the solution milky, because it reacts to form insoluble calcium carbonate
 - $\text{Ca(OH)}_2\text{(aq)} + \text{CO}_2\text{(g)} \rightarrow \text{CaCO}_3\text{(s)} + \text{H}_2\text{O(l)}$

Reactions that make cycad fruit safe to eat

- Type of palm grown in northern and north-eastern Australia
- Traditional source of food for Aboriginal and Torres Strait Islander Peoples for thousands of years, (although is poisonous as harvested)
 - Developed methods of detoxifying the fruit so that it is edible.

1) Physical separation method

- The fruit is cut up, put into coarse mesh bags and soaked in running water for 3–5 days during which time the toxic materials are leached out

- Leaves an insoluble starchy material

2) Heating the cut-up fruit

- Decomposes the toxins first, using a much shorter leaching time to remove any remaining toxins

3) Fruit is immersed in a container or pit of water for several days

- Allows for fermentation (a chemical reaction to destroy the toxins, then the seeds can be washed and used). The



toxins were soluble in water.	
Decomposition of carbohydrate $C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$	

Detoxification of Black Beans (atomi)

- Moreton Bay Chestnut (indigenous plant)
- Seeds are collected and heated in a fire
- Seeds scraped for higher surface area
- Seed mash is placed in a bag and placed in a stream for a long period of time, allowing the water soluble toxins to leach out
- Relies on physical and chemical processes.

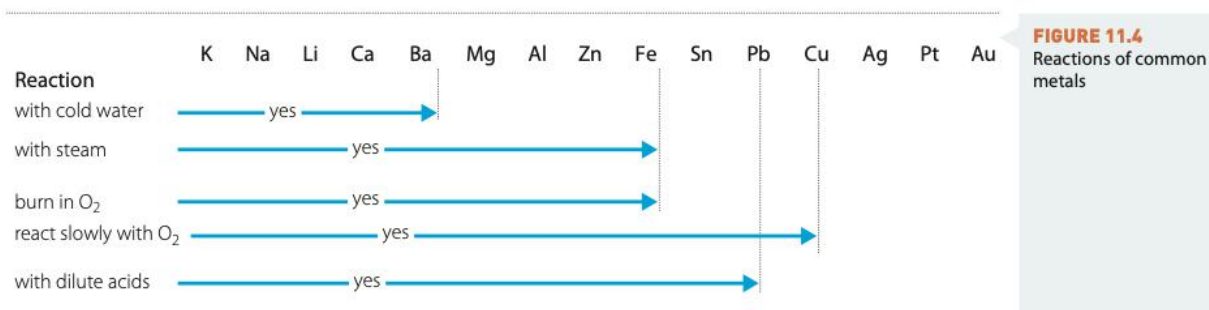
Activity Series for Metals

- Common reactions of metals are with:
 - oxygen (to form oxides)
 - water (to form hydrogen gas)
 - dilute acids (also to form hydrogen gas)
 - cations of other metals (displacement reactions)
 - A displacement reaction is a reaction in which a metal converts the ion of another metal to the neutral atom

Practicals

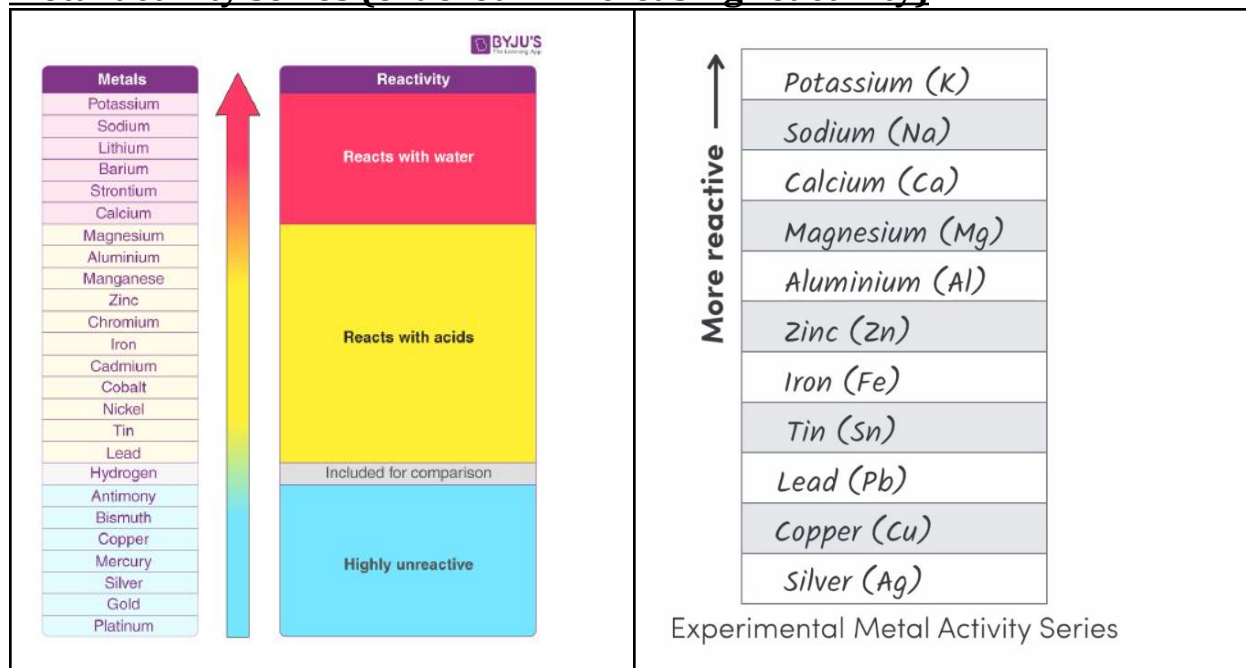
Metal	Oxygen	Water	Acid
Potassium	Reacts readily	N/A	Reacts with cold water
Sodium	Reacts readily	N/A	Reacts with cold water
Calcium	Reacts readily	N/A	Reacts with cold water
Magnesium	Reacts readily	React vigorously in cold acid	Reacts with hot water
Aluminium	Reacts slowly when heated in oxygen	React vigorously in cold acid	No reaction
Zinc	Reacts slowly when heated in oxygen	React vigorously in cold acid	No reaction
Iron	Reacts slowly when heated in oxygen	Reacts slowly in cold acid	No reaction
Tin	Reacts slowly when heated in oxygen	React very slowly in warm acid	No reaction
Lead	Reacts slowly when heated in oxygen	React very slowly in warm acid	No reaction

Copper	Reacts slowly when heated in oxygen	No reaction	No reaction
Silver	No reaction	No reaction	No reaction



→ Copper doesn't react with HCl so it doesn't produce hydrogen gas when it reacts with it.

Metal activity series (ordered in increasing reactivity)



Limitations

- Relies on qualitative observations
- Conditions are not controlled
- Contaminations

Standard reduction potential

- Metals react via oxidation (equation is flipped and sign is flipped)
- Reactivity increases as the list goes up (Potassium is the most reactive)

Metal activity

- A more reactive metal loses electrons more readily and react more vigorously
- Highly reactive metals are rarely found in their elemental state because they easily chemically combine with other elements
- Barely reactive metals are usually found in their elemental state (E.g. gold)

Reactivity with water

- Group 1 (Alkali metals) react vigorously with water to form hydrogen gas and a solution
- Group 2 (Alkaline earth metals) react with water but less vigorously than group 1 to form hydrogen gas and an alkaline solution
 - Beryllium and Magnesium need steam to react vigorously

<u>Down a Group</u>	<u>Across a Period</u>
Metal activity increases down a group	Metal activity decreases across a period
Because: First ionisation energy ↓, atomic radius ↑ and electronegativity ↓.	First ionisation energy ↑, atomic radius ↓ and electronegativity ↑.

Exceptions → Magnesium and Zinc are more reactive than their ionisation energies imply.

Ionisation energy

- Amount of energy required to remove an electron
- As ionisation energy decreases, metal activity increases

Electronegativity

- The ability of atoms of an element to attract a bonding pair of electrons
- As electronegativity decreases, metal activity increases

Atomic radius

- Distance between nucleus and valence electrons
- As atomic radius increases, metal activity increases

Non-Metal Reactivity

Non-metals want to gain electrons, therefore, their trend in reactivity is the exact opposite of those of metals (who want to lose electrons).

<u>Down a Group</u>	<u>Across a Period</u>
Metal activity decreases down a group.	Metal activity increases across a period.
Because: First ionisation energy ↓, atomic radius ↑ and electronegativity ↓. → Hence, electrons are pulled more weakly to the nucleus. → Hence, reactivity of the atom decreases if it's a non-metal.	Across a period, the number of valence electrons approaches eight → hence an increased tendency to gain more electrons, hence non-metal reactivity increases. - Electronegativity increases - Hence non-metal reactivity increases

Redox reactions

- **Redox reaction:** electron transfer reaction
 - E.g. metal with dilute acid to form hydrogen gas
- **Oxidation:** loss of electrons

- gain of oxygen or loss of hydrogen
- **Reduction:** gain of electrons
 - loss of oxygen or gain of hydrogen
- Occur simultaneously because there can be no overall gain or loss of electrons
- OILRIG

Half equations

- Oxidation and reduction processes separately in terms of electrons lost or gained
- To combine them you have to balance the number of electrons

$\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^- \quad \dots (11.11)$ $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2 \quad \dots (11.12)$	Multiply 11.11 by 2 Multiple 11.12 by 3 $2\text{Al(s)} + 6\text{H}^+(\text{aq}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{H}_2(\text{g})$
---	---

Oxidants and reductants

- An oxidant/oxidising agent is a substance that brings about oxidation of another substance
 - Oxidising agent gets reduced
- A reductant/reducing agent is a substance that brings about reduction of another substance
 - Reducing agent gets oxidised

Oxidant = oxidising agent = substance being reduced

Reductant = reducing agent = substance being oxidised

- Better oxidising/reducing agent if magnitude of electromotive force (EMF) is larger.
- When comparing two reduction half reactions, the one with the algebraically larger ϵ goes as written and drives the other one in the reverse.
- The more negative the electrode potential the more reactive the metal.
- The greater the value of ϵ , the greater the oxidising strength of the oxidised form (left-hand side) of the half reaction.
- The (algebraically) smaller the value of ϵ , the greater the reducing strength of the reduced form (right-hand side) of the reduction half reaction.

Oxidation numbers (states)

- The oxidation number/state of an element in a molecule or ion is the charge the atom of that element would carry if the molecule or ion were completely ionic
- The sum of the oxidation numbers of all the elements in a species (molecule or polyatomic ion) must equal the net charge on the species
- Usually, valency = oxidation state (E.g. Cu in Cu_2S has a valency of 1 = Cu is in the +1 oxidation state)
- **Increase in oxidation state (more positive) = loss of electrons = oxidation**
- **Decrease in oxidation state (more negative) = gain of electrons = reduction**
- The coefficients in a balanced equation play no part in determining oxidation numbers

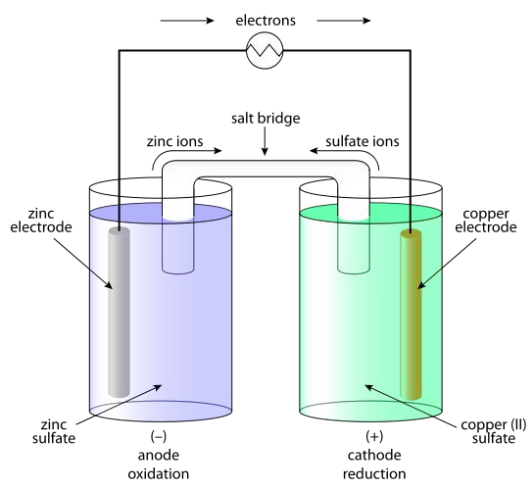
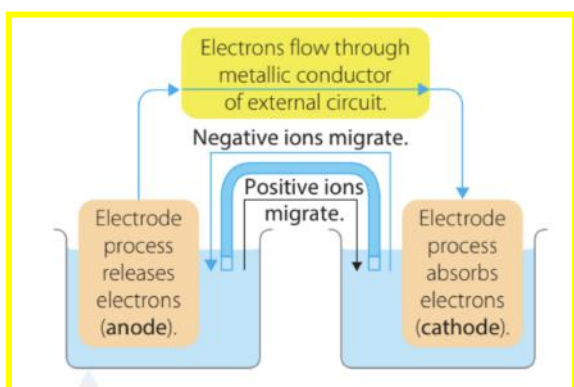
Rules for oxidation state

1. A substance present in the **elemental state** is assigned an oxidation number of **zero**, regardless of the formula of the molecule of the element.

- a. Copper, silver and iron exist as discrete atoms; oxygen, hydrogen and chlorine exist as diatomic molecules; phosphorus exists as P_4 and sulphur as S_8 . Regardless of their structures or formulae, all the elements are considered as having an oxidation number of zero.
 - b. Other rules refer to elements in compounds.
2. The oxidation number of a monatomic ion is simply the charge on the ion.
 3. Fluorine always has an oxidation number of -1 .
 4. Hydrogen has an oxidation number of $+1$ except where rule 2 requires it to be -1 (such as in NaH).
 5. Oxygen has an oxidation number of -2 unless rules 2, 3 and 4 dictate that it have a different value (such as $+2$ in F_2O and -1 in hydrogen peroxide, H_2O_2)
 6. Chlorine, bromine and iodine have oxidation numbers of -1 unless rule 5 requires them to have a different value
 7. With these six rules as a starting point, it is possible to work out oxidation numbers of other atoms from the requirement that:
 8. The algebraic sum of the oxidation numbers of the elements in a molecule or ion must equal the net charge on the species

Galvanic cells

- **Electrochemistry:** The area of science about chemistry and electricity
- **Galvanic cell:** device in which a chemical reaction occurs and generates electricity



Half cells: Each

Electrolyte: substance that conducts electricity in solution /molten state

Electrode: Conductor that connects the external electrical circuit to the galvanic cell

Anode: Where oxidation occurs

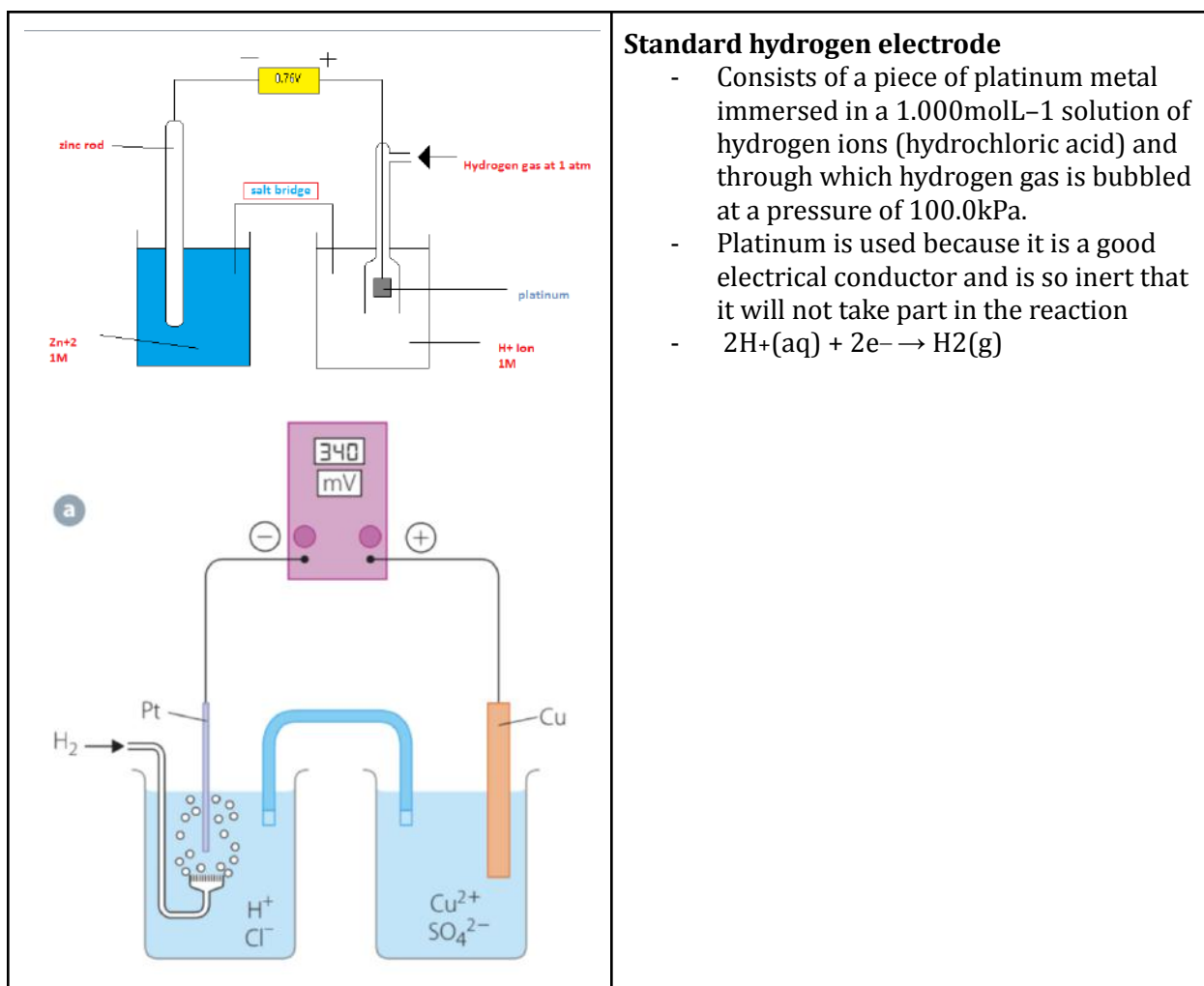
Cathode: Where reduction occurs

Wire: Allow electrons to flow from anode to cathode

Voltmeter:

Salt bridge: Provides electrical contact between two solutions in a galvanic cell

- Prevents the build up of negative charge in the cathode (happens because cathode keeps gaining electrons)
- Prevents the build up of positive charge in the anode
- Allows the migration of ions between two electrolytes in a galvanic cell to maintain electrical neutrality
- The electrolyte used in the salt bridge must be one that does not react with any of the ions in the two solutions it is connecting (E.g. potassium nitrate)



Standard hydrogen electrode

- Consists of a piece of platinum metal immersed in a 1.000molL⁻¹ solution of hydrogen ions (hydrochloric acid) and through which hydrogen gas is bubbled at a pressure of 100.0kPa.
- Platinum is used because it is a good electrical conductor and is so inert that it will not take part in the reaction
- $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$

- The **electromotive force (EMF)** of a galvanic cell is the potential difference (voltage) across the electrodes of the cell when a negligibly small current is being drawn. It is the **maximum voltage** that the cell can deliver.
- **Standard electrode potentials** are ON DATA SHEET
 - Reduction equations so flip equation and sign for oxidation
 - Top right = Strongest reducing agent (gets oxidised easiest)
 - Bottom left = Strongest oxidising agent (gets reduced easiest)
 - Calculating voltage for redox reactions:

$$E_{\text{total}}^{\ominus} = E_{\text{red}}^{\ominus} + E_{\text{oxid}}^{\ominus}$$

- Allows us to determine if redox reactions are written and spontaneous
 - **Negative total voltage** = **nonspontaneous** reaction and must be reversed
 - **Positive total voltage** = **spontaneous** reaction and is written
- For metals, reactivity means the ease with which the metal can be oxidised (the ease of an oxidation half reaction to occur)
 - The more negative the electrode potential = the more reactive the metal
 - More reactive metal displaces the less reactive metal

- Less reactive metal deposits on the more reactive metal

Salt Bridge

1. Completes the external circuit
2. Prevents charge build up to ensure that the redox reaction continues through the movement of the ions in the salt bridge.
 - a. In the external circuit electrons are moving from the anode to the cathode, therefore:
 - i. Cations move to the cathode
 - ii. Anions move to the anode
3. Is soaked in KNO_3 or NaNO_3 .

Observations in Galvanic Cell Reactions

1. Change in mass of electrodes $\rightarrow$ decrease in anode, increase in cathode
2. Change in concentration of ions in beakers
 - a. Reflected in change in colour intensity of solution, (less reactive metal solution loses its colour) e.g copper ions (blue-green)
3. Voltmeter reads an EMF

Electrode Reaction	Electrode	Polarity/Sign of Electrode
Oxidation	Anode	Negative
Reduction	Cathode	Positive

Conditions to Measure Standard Electrode Potential

1. A half-cell with an electrode in 1.0M solution of its ions (at a pressure of 100.0kPa).
2. Connection to standard hydrogen electrode.
 - a. **Reduction process occurring in standard hydrogen electrode:**
 - i. $2\text{H}^+_{(aq)} + 2e^- \rightarrow \text{H}_{2(g)}$
3. Voltmeter between two half-cells to measure potential difference.

The more positive the value for E^0 the greater the driving force for reduction.

$\rightarrow$ This is just talking about E^0 as written on the data sheet, its not E^0_{total} .

- $E^0_{total} = E^0_{reduction} + E^0_{oxidation}$ (the sign of these are important)
- If $E^0_{total} > 0$, the reaction is spontaneous.
- If $E^0_{total} < 0$, the reaction is not spontaneous.

Types of Galvanic Cells

1. Two metal electrodes
2. An inert conductor (platinum) due to the presence of ions on both sides of the reaction.
 - a. Fe^{2+} and Fe^{3+} ions
3. Gas bubbling into the solution with an inert conductor (platinum)
 - a. Chlorine, chlorine electrode
 - b. Hydrogen, hydrogen ion electrode

$\rightarrow$ Check **Standard Hydrogen Electrode** section

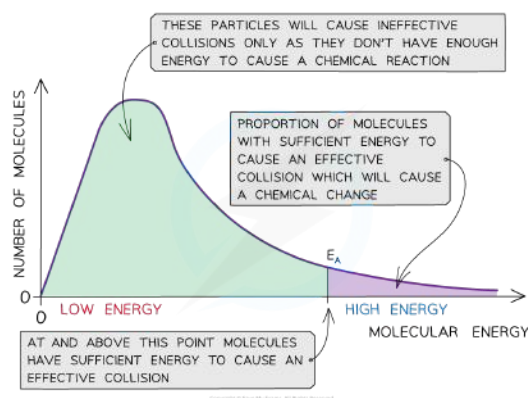
Rates of reaction

- Refers to the rate at which reactants are used up/products form in a chemical reaction
- Average rate = change in concentration/time taken
- mols/sec

Collision theory

1. A chemical reaction can only occur when the reactant particles **collide** and are moving **fast** enough.
2. Only a certain percentage of collisions will result in a **successful** reaction.
3. The particles need to collide with enough kinetic **energy** to break existing bonds (Maxwell-Boltzman curve)
4. Collision must occur in the **correct orientation** otherwise particles will bounce off each other (no reaction)

The frequency of successful reactions determines the rate of reaction



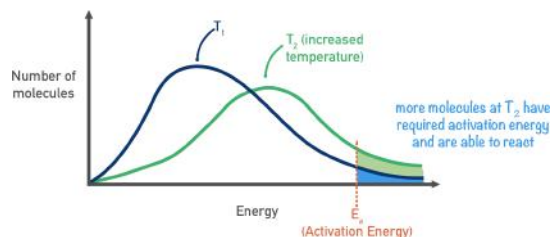
Maxwell boltzmann diagram

Collision theory states that for a chemical reaction to occur, reactant particles must collide with one another. In order for collisions to be successful, reacting particles must collide with sufficient kinetic energy and with the correct orientation. **The rate of reaction depends on the frequency of these successful collisions.**

Factors influencing the rate of reaction

1. Temperature

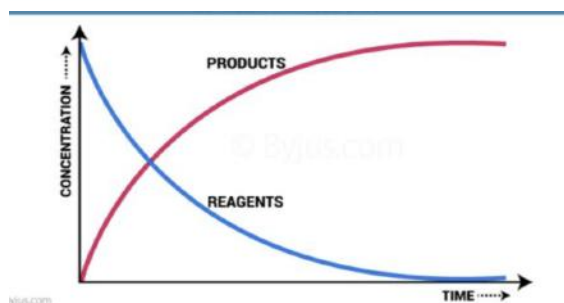
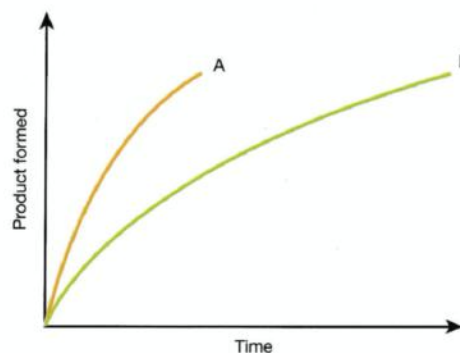
Increasing temperature increases the average kinetic energy of particles. This also increases the strength of particle collision and the amount of reactant particles with a greater kinetic energy than activation energy. Thus increasing the frequency of particle collisions and the amount of successful collisions.



2. Concentration

Increasing the concentration increases the number of particles which increases the frequency of collisions and the amount of successful collisions, thus **increasing** the rate of reaction. However, as the **reaction progresses**, the concentration of the reactants **decreases** because they are being used up to form the products. This reduces the number of successful collisions by reactant particles, which slows down the rate of the reaction.

- **Pressure** (Increase rate of reaction by increasing pressure/decreasing volume because particles will be closer together; increasing frequency of collisions)
- **Stirring (Heterogeneous reactions)** continuously brings up fresh solutions to keep the solid particles dispersed instead of letting them sink to the bottom



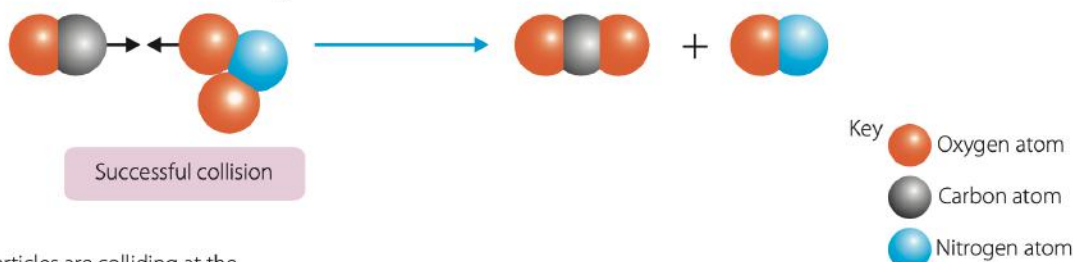
→ **Rate of reaction generally decreases as a reaction proceeds (because concentration of reactants decreases).**

Concentration-versus-time graph

- 1) The rate of reaction at any particular time is the gradient of the curve at that time
- 2) ALWAYS POSITIVE = the rate of the reaction is the magnitude of the gradient
- 3) As the reaction progresses, the rate of reaction decreases as reactants are used up
- 4) Steeper gradient = higher rate of reaction

3. Orientation

Particles are colliding at the right orientation and with sufficient energy.



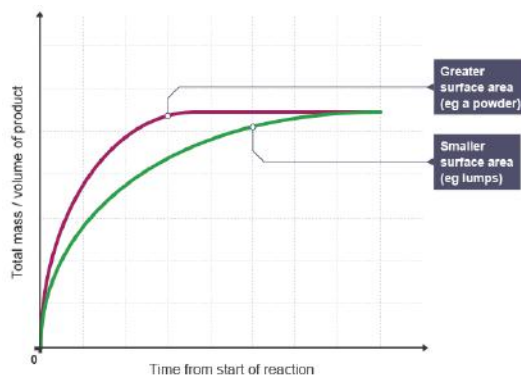
Particles are colliding at the wrong orientation.



FIGURE 13.8 Correct particle orientation is required for reaction to occur.

4. Surface area (Heterogeneous reactions)

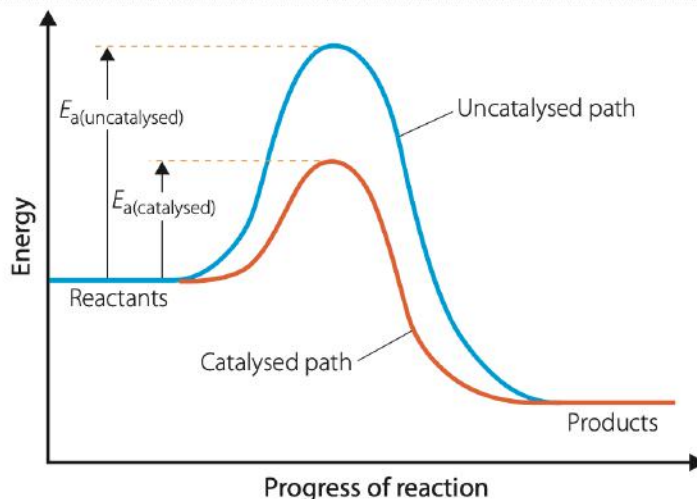
Increasing the surface area of solid reactants increases the exposed surface where more particles can react which increases the frequency of particle collisions and the amount of successful collisions, thus increasing rate of reaction.



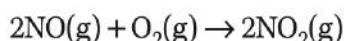
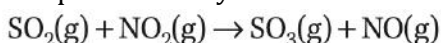
5. Catalysts

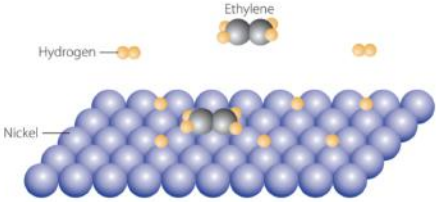
- 1) Provides an **alternate pathway** with a **lower activation energy**
 - a) Reaction happens faster
 - b) Reactant particles are able to react at a lower activation energy which means more particles can collide and react
 - 2) Orient molecules **correctly** and/or bring them in close **proximity** to allow them to react successfully
 - 3) Form an **intermediate complex**
 - 4) Don't affect the yield of the reaction
- It is a catalyst as long as:
- **Isn't** used up/consumed in the reaction (is in the reactants and the products)
 - **Same** amount of mass at the start and end of the reaction (same yield)
 - Activation energy is lower with a catalyst
 - Enthalpy is the same

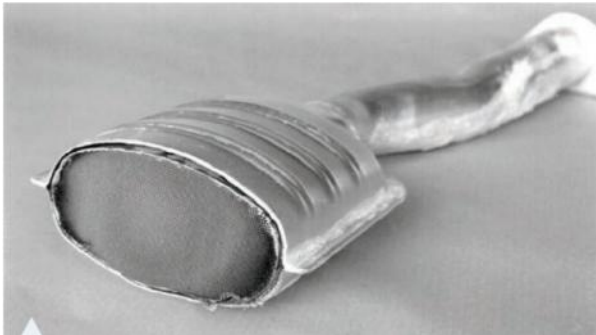
FIGURE 13.9 Energy profiles for catalysed and uncatalysed paths of a reaction



Example of a Catalysed reaction:



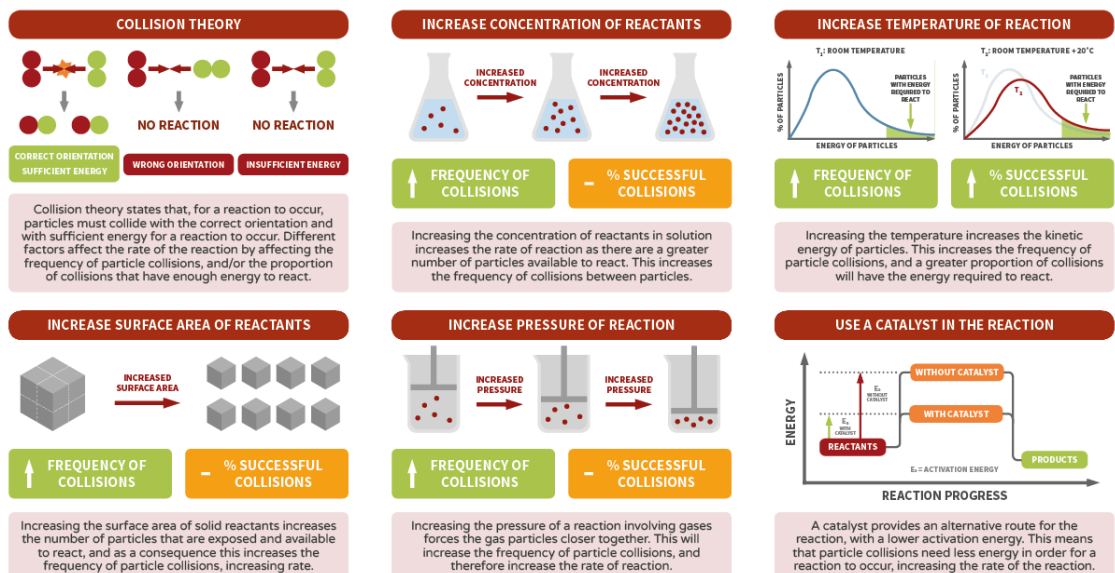
<u>Heterogeneous catalyst</u>	<u>Homogeneous catalysts</u>
Provide a surface for the reactant particles to stick onto, weakening the intramolecular forces and lowering the E_a so that more particles meet the required energy conditions and the reaction occurs at a faster rate. (Surface Area can increase $\rightarrow$ usually solid)	Catalyst is in the same state as the reactants and is regenerated at the end of the reaction It reacts with one of the reactants to form an intermediate complex that reacts with another reactant to form a product.
Hydrogenation of ethylene to form ethane is catalysed by nickel - Allows ethylene to adhere to the surface (bringing particles closer together) - Facilitates breaking of the double bond  $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$	Overall reaction : $\text{A} + \text{B} \rightarrow \text{P}$ $[\text{A}] + \text{catalyst} \rightarrow \text{Intermediate}$ $\text{Intermediate} + [\text{B}] \rightarrow [\text{P}]$ $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$ - Very slow reaction (has a high activation energy) (NO_2 makes the reactions occur rapidly) 1. $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$ 2. $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$ - No overall consumption of NO_2
Catalytic exhausts of modern cars. 1) Rhodium catalyses the reaction between carbon monoxide and nitric oxide to form carbon dioxide and nitrogen gas	Enzymes - Biological catalysts and proteins that allow important reactions to happen in all living cells

2) Platinum catalyses the reactions of carbon monoxide and unburnt hydrocarbon to carbon dioxide - Removing the three most harmful pollutants from the exhaust gas - Deposited as very thin films on all the surfaces of a ceramic honeycomb-shaped block - Allows for the catalysis to occur on a large surface (increasing SA) and for the production of less toxic gases 	- E.g. In the saliva amylase combines with starch (using intermolecular forces) and breaks it down into glucose/maltose - Create new compounds by bringing particles closer together 14 Catalytic converters (exhausts) in cars have two catalysts. rhodium catalyses reaction i and platinum catalyses reaction ii. i $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ ii $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$ Platinum also catalyses: hydrocarbon + $\text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ a In the design of the converter propose a reason for making reaction i considerably faster than ii. b Identify some factors that could be varied to control the relative rates of these two reactions. a. (i) has to be faster than (ii) because it needs CO to react with NO. If reaction (i) occurs to completion, then remaining CO will just react, however, if all the CO is used up in (ii) then there's unreacted NO which is a harmful pollutant. b. Temperature, SA of catalyst, etc.
--	---

	Heterogeneous catalyst	Homogeneous catalysts
Information	- Different state as reactants - Reactant particles absorb - Acts as a surface for the reactant molecules to align (assists with orientation), it also weakens (or breaks) reactant bonds (decreases E_A).	- Same state as reactants - Reacts with the reactants to form an intermediate complex that reacts with another reactant to form a product. - Regenerated at the end of the reaction.
Advantages	- Can easily be removed from the reactants.	- High SA in contact with reactants so catalyst has a big impact in increasing ROR.
Disadvantages	- Catalyst poisoning - Doesn't increase ROR as much as a homogeneous catalyst.	- Hard to separate from the reactants.

MAKING CHEMICAL REACTIONS HAPPEN FASTER

There are a number of different things that we can change to make a chemical reaction faster. Here, we explain the concept of collision theory, and how it can be used to explain the effects of five different factors on the rate of a chemical reaction.



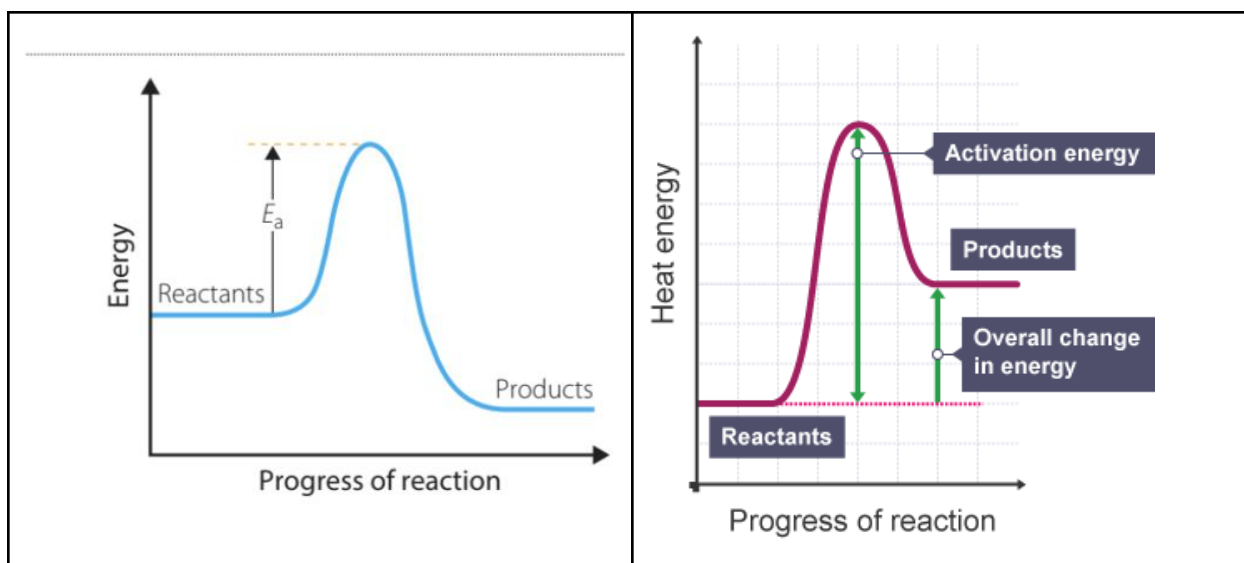
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→ The Haber Process combines hydrogen and nitrogen to make ammonia using an iron catalyst. This is another reversible reaction, and another example of heterogeneous catalysis.

Activation energy

- The energy barrier between reactants and products
- Minimum amount of energy particles must collide with in order for a reaction to occur
- The higher the activation energy, the harder it is for reactants to overcome it and form products and so the slower the reaction rate (at a given temperature)
- The lower the activation energy, the higher the rate of reaction (because more particles are above E_A , hence more particles have sufficient energy to react, hence ROR is higher).



MODULE 4- Drivers of Reactions

Energy Changes in Chemical Reactions

Inquiry question: What energy changes occur in chemical reactions?

- Conduct practical investigations to measure temperature changes in examples of endothermic and exothermic reactions, including:
 - combustion
 - dissociation of ionic substances in aqueous solution

- Investigate enthalpy changes in reactions using calorimetry and $q = mc\Delta T$ (heat capacity formula) to calculate, analyse and compare experimental results with reliable secondary-sourced data, and to explain any differences
- construct energy profile diagrams to represent and analyse the enthalpy changes and activation energy associated with a chemical reaction
- model and analyse the role of catalysts in reactions

Enthalpy and Hess's Law

Inquiry question: How much energy does it take to break bonds, and how much is released when bonds are formed?

- Explain the enthalpy changes in a reaction in terms of breaking and reforming bonds, and relate this to the law of conservation of energy
- Investigate Hess's Law in quantifying the enthalpy change for a stepped reaction using standard enthalpy change data and bond energy data, for example:
 - carbon reacting with oxygen to form carbon dioxide via carbon monoxide

- Apply Hess's Law to simple energy cycles and solve problems to quantify enthalpy changes within reactions, including but not limited to:

- heat of combustion
- enthalpy changes involved in photosynthesis
- enthalpy changes involved in respiration

Entropy and Gibbs Free Energy

Inquiry question: How can enthalpy and entropy be used to explain reaction spontaneity?

- Analyse the differences between entropy and enthalpy
- Use modelling to illustrate entropy changes in reactions
- Predict entropy changes from balanced chemical reactions to classify as increasing or decreasing entropy

- Explain reaction spontaneity using terminology, including:
 - Gibbs free energy
 - enthalpy
 - entropy

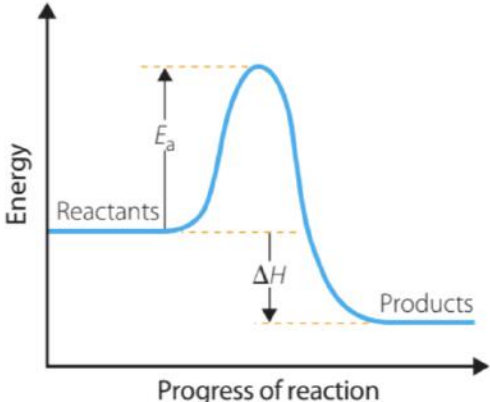
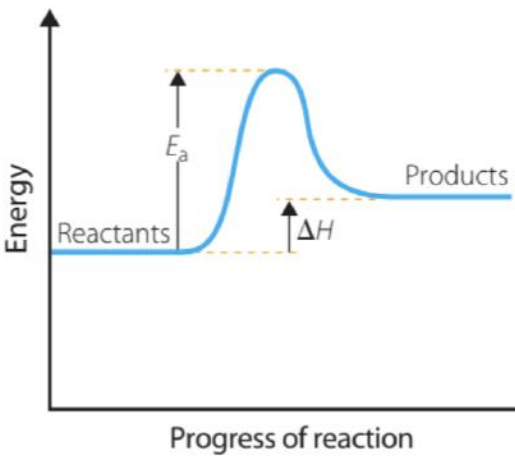
- Solve problems using standard references and $\Delta G_o = \Delta H_o - T\Delta S_o$ (Gibbs free energy formula) to classify reactions as spontaneous or nonspontaneous

- Predict the effect of temperature changes on spontaneity

Energy Changes in Chemical Reactions

- **Thermochemistry:** Study of energy changes in chemical reactions

<u>Exothermic</u>	<u>Endothermic</u>
- Release heat - The test tube gets hot - decrease in chemical energy and the 'lost' chemical energy is released as heat, which warms up the test tube	- Absorbs heat - Test tube gets cold - requires heat which it can only get from the test tube to convert to chemical energy so the test tube gets cold
- combustion of coal and petrol - synthesis reactions (or direct combination reactions) - reactions of metals with water & acids - reactions of acids with bases and carbonates - dissolutions of salts in water (calcium chloride and sodium carbonate) - Respiration	- decomposition of copper nitrate or calcium carbonate - some precipitation reactions such as $\text{MgCl}_2 + \text{Na}_2\text{CO}_3$ - photosynthesis - dissolutions of salts in water (potassium sulphate and ammonium chloride) - More exothermic reactions than endo

Exothermic reaction 	Endothermic reaction 
- ΔH is negative - q is negative	- ΔH is positive - q is positive
Examples: - Combustion - Cell respiration - Epsom salts and water - Yeast and hydrogen peroxide	Examples: - Photosynthesis - Acetic acid and sodium bicarbonate

Energy profile diagrams

1. Energy of reactants
 2. Energy of products
 3. Change in enthalpy
 4. Activation energy (reactants $\rightarrow$ peak of curve)
- Fill the whole page or use at least $\frac{1}{3}$ of the page

Quantity of heat

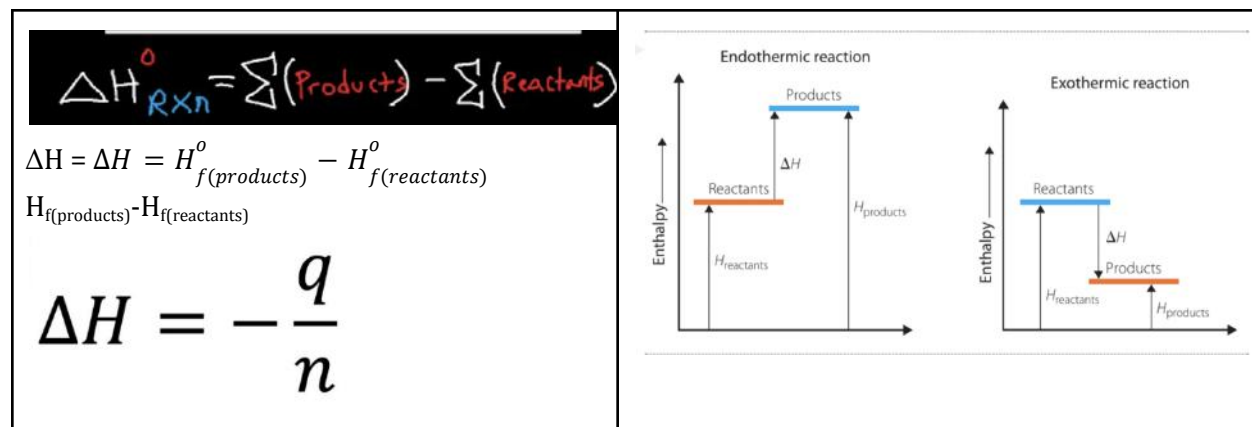
- **Calorimetry:** The measurement and calculation of heat changes
- Calorimeter: Polystyrene cup
 - Minimises heat fluctuation as it is an insulating material
 - Light weight material so it doesn't absorb as much heat from the water
 - Water surrounds the flame so it absorbs as much of the heat of the flame as possible
 - Forming soot doesn't release as much energy that can heat water, therefore incomplete combustion reduces the energy output
- Amount of heat or quantity of heat is different from temperature
- Amount of heat energy is proportional to the mass of the substance involved
- The **specific heat capacity** (c) of a substance is the amount of heat required to increase the temperature of the unit mass of the substance by exactly 1°C (or through 1 kelvin)
 - Measured in J/K/g

$$q = mc\Delta T$$

- $\Delta T = T_f - T_i$
- If the temperature increases, ΔT is positive and therefore reaction is exothermic
 - This means that the object has lost heat
- If the temperature decreases, ΔT is negative and therefore reaction is endothermic
 - This means that the object has gained heat

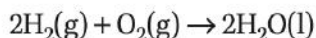
Enthalpy (H)

- Sometimes referred to as the **heat of reaction**.
- The change in enthalpy (ΔH) is defined as the heat absorbed **per mole** of specified reactant or product when the reaction occurs at constant pressure (kJ/mol)
- **Molar enthalpy of solution or heat of solution (ΔH_{soln})** = the heat absorbed when one mole of the substance dissolves in a large excess of water
 - For endothermic reactions, heat absorbed, ΔH is positive
 - For exothermic reactions, heat released, ΔH is negative
- **Dissolution:** The process of a solute becoming aqueous (dissolving).
 - Is an endothermic reaction as there is a slight energy difference between a substance in its solid and aqueous state
 - Only applies to soluble molecules that are not ionic because the molecule doesn't break apart as ionic substances do.
 - Either exothermic or endothermic
- **Dissociation:** Dissolving ionic substances in aqueous solution to form separated ions
- Typically endothermic?

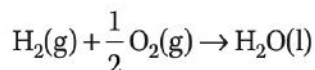


Enthalpy (H) - Ambiguity

For example, the statement that ΔH for the formation of water from hydrogen and oxygen is -572 kJ is ambiguous because it does not make clear whether the reaction concerned is:



or



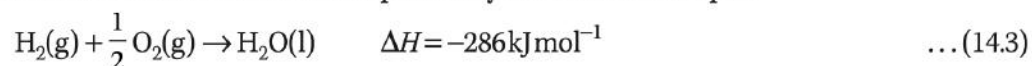
Ambiguity is removed by writing:

$$\Delta H = -572 \text{ kJ per mole of oxygen}$$

or

$$\Delta H = -286 \text{ kJ per mole of water (or per mole of hydrogen)}$$

Alternatively, the reaction referred to can be specifically written; for example:



Here, 'per mole' means 'per mole of the reaction *as written*', in this case 'per mole of hydrogen' or 'per mole of water'.

We put 'per mole' in the units of ΔH to show that the value is not a 'per molecule' one, because, as we saw in section 7.12, equations can be read in terms of molecules or moles.

Changes of state, for example from solid to liquid to gas and vice-versa, involve absorption or release of energy. Hence changes in ΔH depend upon the physical state (solid, liquid, gas or solution) of the reactants and products, and therefore in chemical energy contexts we must *always* indicate the physical state of the substances involved in chemical equations.

Heat of Dissolution

Equation looks like this: $\text{C}_6\text{H}_{12}\text{O}_6(\text{s}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6(\text{aq})$ or $\text{KNO}_3(\text{s}) \rightarrow \text{K}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$.

$\Delta H_{\text{soln}} < 0$ — reaction is exothermic	$\Delta H_{\text{soln}} > 0$ — reaction is endothermic
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Heat of combustion

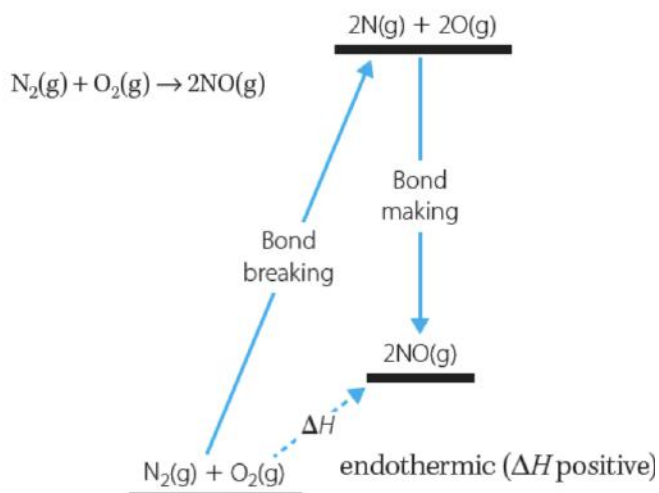
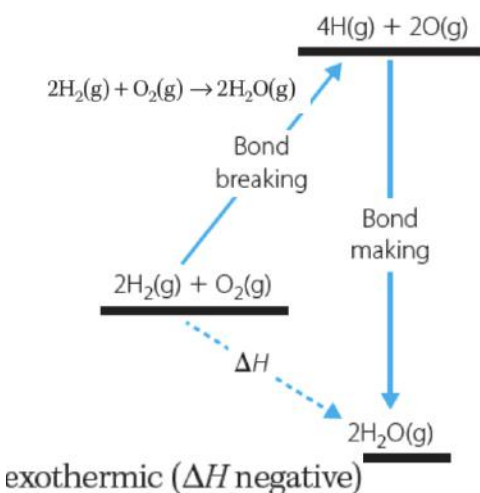
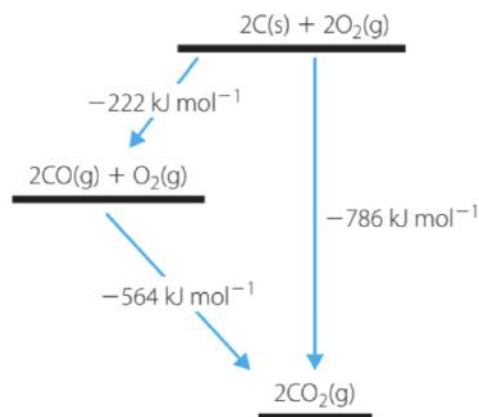
- **Molar heat of combustion:** the heat liberated when 1 mole of a substance undergoes complete combustion with oxygen at a constant pressure of 100.0kPa, with the final products being carbon dioxide gas and **liquid** water.
- Combustion is **exothermic** therefore enthalpy change is negative
 - molar heat of combustion (= heat released) is positive
 - enthalpy change for a combustion process and enthalpy of combustion (= heat absorbed) are negative **because heat is released not absorbed (!!!!)**.
- The heat of combustion is measured indirectly by having the combustion reaction heat another substance (water)
 - The heat absorbed by a known mass of water that has been heated through a measured temperature increase is calculated
 - The law of conservation of energy states that energy can be neither created nor destroyed but it can be changed from one form to another
 - heat released by the combustion reaction = heat absorbed by the water

- Theoretical value can differ from experimental value due to incomplete combustion or heat loss to surroundings

Hess's Law (only applicable if given heat of formation values)

- Based on the law of conservation of energy.
- Enthalpy change is independent of the path taken.
- Hess's law states that for a chemical reaction the enthalpy change in going from reactants to products is constant, regardless of what set of reaction steps is used to bring it about
- A reaction that forms a product in multiple steps will have the same enthalpy as the whole reaction
- Used to calculate the enthalpy change for a reaction from ΔH values of other reactions
 - If a reaction is reversed, the sign of ΔH is changed
 - If a reaction is doubled (or halved) then ΔH is doubled (or halved)
 - If reagents are on the reactant and product side and have the same coefficient, they can be cancelled out
- $H_1 = H_2 + H_3$

1. Draw horizontal lines representing the initial state, intermediate state(s) and final state.
2. Put all the reactants on the initial state line and all the products on the final state line and relevant substances on the intermediate one(s).
3. Draw arrows showing changes from one state to another and put the symbols or values of the enthalpy changes on the arrows.



- Time is not an appropriate x-axis label because both the forwards and reverse reactions

can occur at the same time.

Standard enthalpy change ($\Delta H^\ominus$)

- The enthalpy change when the reaction occurs with all reactants and products present in their standard states
 - For a pure substance, the standard state is the stable form of the substance (solid, liquid or gas) at a pressure of 100.0kPa and at the specified temperature.
 - For a substance in a gaseous mixture, the standard state is that substance present at a pressure of 100.0kPa.
 - For a substance in solution, the standard state is that substance present at a concentration of exactly 1 mol/L

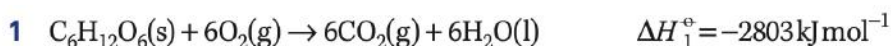
Standard enthalpy of formation ($\Delta H_f^\ominus$)

- Increase in enthalpy when one mole of a compound in its **standard state** is formed from its elements in their standard states
- The standard enthalpy of formation of an **element** in its **standard state** is zero (!!!).

$$\Delta H^\ominus = \sum \Delta H_f^\ominus (\text{products}) - \sum \Delta H_f^\ominus (\text{reactants})$$

- State of water should be a liquid in **heat of combustion** reactions (not gas as observed in real life).

Photosynthesis	(Cellular) Respiration
$6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) + 6\text{O}_2(\text{g})$	$\text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) + 6\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$
Endothermic & nonspontaneous	Exothermic & spontaneous

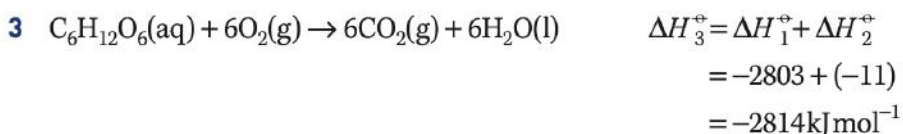


(Enthalpy change is the negative of heat of combustion.)



(Reaction 2 is the reverse of dissolution so its $\Delta H^\ominus$ is reversed also)

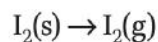
Adding equations 1 and 2 and cancelling solid glucose gives:



Hence the enthalpy change for the respiration reaction in equation 15.9 is $-2814 \text{ kJ mol}^{-1}$, and so the enthalpy change for the photosynthesis reaction in equation 15.8 is $+2814 \text{ kJ mol}^{-1}$.

This discussion has illustrated how Hess's law can help us calculate a desired but difficult-to-measure quantity from other more easily measurable quantities.

However, chemists do talk about the standard enthalpy of formation of elements not in their standard states. For example the standard enthalpy of formation of gaseous iodine at 298K, $\Delta H_f^\ominus (\text{I}_2, \text{g})$ is the enthalpy change for:



It is 62 kJ mol^{-1} . It is actually the heat of sublimation of iodine. If an element is required to be in a state other than its standard state at the specified temperature, the state must be specifically mentioned. The standard enthalpy of formation of bromine at 298K is for liquid bromine. If we want the value for bromine as a gas, this must be clearly mentioned.

Heat of Atomisation

- The heat of atomisation, ΔH_{at} , of an element is the enthalpy change for the production of gaseous atoms from the element in its standard state at a pressure of 100.0kPa and at the temperature in question.
- It is usually expressed per mole of atoms.
- Always positive
- E.g. $\text{C}(\text{s}) \rightarrow \text{C}(\text{g})$, $\frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$

Bond energy

- **Energy** is needed to **break chemical bonds** and **energy is released** when new chemical bonds are **formed**
- It is measured in kilojoules per mole of bonds and is always positive

$$\Delta H = \{\text{energy required to break bonds}\} - \{\text{energy released when new bonds form}\}$$

- $\Delta H = \{\text{energy of reactants}\} - \{\text{energy of products}\}$
- Exothermic: energy released is greater than energy required to break bonds (-)
- Endothermic: energy required to break bonds is greater than energy released (+)
- The greater the bond energy = the more stable the compound is and the less reactive it is
- The bond enthalpy refers to the energy required to break one mole of the bonds in a substance's gaseous state

Average Bond Enthalpies (kJ/mol)

Single Bonds

C—H	413	N—H	391	O—H	463	F—F	155
C—C	348	N—N	163	O—O	146		
C—N	293	N—O	201	O—F	190	Cl—F	253
C—O	358	N—F	272	O—Cl	203	Cl—Cl	242
C—F	485	N—Cl	200	O—I	234		
C—Cl	328	N—Br	243			Br—F	237
C—Br	276			S—H	339	Br—Cl	218
C—I	240	H—H	436	S—F	327	Br—Br	193
C—S	259	H—F	567	S—Cl	253		
		H—Cl	431	S—Br	218	I—Cl	208
Si—H	323	H—Br	366	S—S	266	I—Br	175
Si—Si	226	H—I	299			I—I	151
Si—C	301						
Si—O	368						

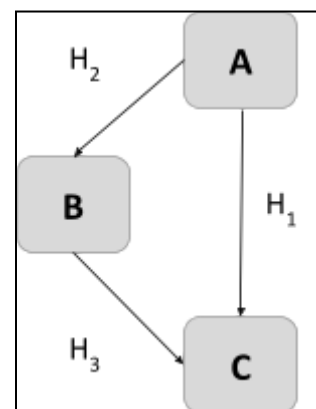
Multiple Bonds

C=C	614	N=N	418	O ₂	495
C≡C	839	N≡N	941		
C=N	615			S=O	523
C≡N	891			S=S	418
C=O	799				
C=O	1072				

Entropy (s)

- Measures the amount of randomness or chaos or lack of ordered structure in a substance
- 1. Moving from an ordered state to a less ordered one corresponds to an increase in entropy
- 2. An increase in the number of gaseous moles corresponds to an increase in entropy
- Standard molar entropy = the entropy of one mole of the substance in its standard state
- Measured in J/K/mol

$$\Delta S^{\ominus} = \sum \Delta S^{\ominus}(\text{products}) - \sum \Delta S^{\ominus}(\text{reactants})$$



Bond Energy, Stability and Reactivity

Relationship between bond energy, stability and reactivity

Now that we have determined some bond energies we are in a position to consider more closely the relationship between bond energy, stability and reactivity. In section 15.1 it was stated that the greater the bond energy the more stable the compound is and the less reactive it is. We can now test that hypothesis against our measured bond energies.

Oxygen, O_2 , is a very stable gas. Its allotrope, ozone, O_3 , is notoriously unstable: samples of ozone liquid often spontaneously explode. This is because of its lack of stability and high reactivity, both arising from the fact that the O—O bonds in ozone have much lower bond energies than the one in O_2 : 302 kJ mol^{-1} in ozone and 498 kJ mol^{-1} in O_2 .

Water, $H-O-H$, is very stable and not particularly reactive. Hydrogen peroxide, $H-O-O-H$, is much less stable and considerably more reactive. This is because of the relatively low bond energy for the O—O bond in hydrogen peroxide (146 kJ mol^{-1} for O—O compared with 463 kJ mol^{-1} for O—H). The two H—O bonds in hydrogen peroxide have similar bond energies to those in water. However the strength of the weakest bond in the molecule is the most important in deciding stability and reactivity.

$$\Delta H^\circ = \{\text{energy required to break bonds}\} - \{\text{energy released when new bonds form}\}$$

Alternatively:

$$\Delta H^\circ = \{\text{sum of bond energies of bonds broken}\} - \{\text{sum of bond energies of bonds formed}\} \dots (15.7)$$

Gibbs free energy

Standard Gibbs free energy change (ΔG): a measure of the net result of the energy and entropy drives for a reaction

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

- $\Delta G < 0$, the reaction is spontaneous
- A spontaneous reaction is one that occurs without any ongoing input of energy (although it may need a small input of energy in the form of a flame or spark to get it started)

Spontaneous at all times $\Delta H = \text{negative} \ \& \ \Delta S = \text{positive}$	Nonspontaneous $\Delta H = \text{positive} \ \& \ \Delta S = \text{negative}$
Spontaneous at low temperatures + enthalpically driven $\Delta H = \text{negative} \ \& \ \Delta S = \text{negative}$	Spontaneous at high temperatures + entropically driven $\Delta H = \text{positive} \ \& \ \Delta S = \text{positive}$